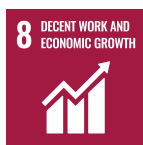




MTP Department
SIMHADRI



Ref: SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

Date:

10/06/2023(17:08:28)

Subject: LMI/Approval for Revision-2 of LMI on Quality Management in Maintenance / Overhauling

SIMHADRI SUPER THERMAL POWER STATION

Maintenance Planning Department

Sub: - Approval for revision-2 of "LMI /OGN/OPS/SYST/018: Quality Management in Maintenance & Overhauling" - Reg

Operation Guidance Note (OGN) on "Quality Management in Maintenance & Overhaul" was revised by COS vide COS-ISO-00-OGN/OPS/SYST/018 in April 2023. Based on this our station LMI on "Quality management in Maintenance & Overhauling" (LMI/OGN/OPS/SYST/018) is revised.

Major changes incorporated in this LMI were given in the below table.

LMI	Revised Sections (Major Changes)
LMI/OGN/OPS/SYST/018 Quality Management in Maintenance & Overhaul	1. Annexure-A: Unit overhauling norms for NTPC thermal units. 2. Annexure-B: Rolling plan for capital overhauling of Steam Turbine (200 MW / 500 MW Units) 3. Annexure-C: Rolling plan for OH/servicing of TG & Aux. equipment's during OH

	4. Annexure-D: Revised Rolling plan for OH of Generators for Coal and Hydro (Koldam) 5. Annexure-E: Record sheet for trial run of motor. 6. Entire C&I QAP for OH in clause 6.27 has been incorporated. 7. Additional boiler checks incorporated in clause 6.1.2 as per OGN. 8. Necessary changes to Electrical system have been incorporated in clause 6.13 9. Checklist of Offsites areas added at clause 6.33
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Enclosures:

1. Revised LMI on "Quality Management in Maintenance & Overhauling".
2. Revised OGN on "Quality Management in Maintenance & Overhauling".

Put up for approval by Competent Authority

Submitted by:

चौ. श्रीवेंकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP) ,Simhadri
(CHSRIVENKATAPHANIKUMAR@NTPC.CO.IN, 9440100534)

On:10/06/2023(17:08:28) **Ref:**SMTTP/Maint. PLNG./2023-24/1035/MTP/203426

checked and forwarded for review

Submitted by:

पेड्डासिंगू नाग सुरेश - उप महाप्रबंधक

Peddasingu Naga Suresh - 008111 (DY. GENERAL MANAGER ,MTP)
(PNSURESH@NTPC.CO.IN, 8332970695) ,Simhadri

On:10/06/2023(17:15:51) **Ref:**SMTTP/Maint. PLNG./2023-24/1035/MTP/203426

Reviewed and forwarded.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:12/06/2023(10:41:04) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:12/06/2023(12:17:37) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Please examine.

Submitted by:

अर्जुन प्रदीप नायर - अपर महाप्रबंधक

Arjun Pradeep Nair - 004419 (ADDL. GENERAL MANAGER ,TECH SERVICES)
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On:13/06/2023(10:32:45) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Pl review

Submitted by:

विमल वाडिलाल शाह - उप महाप्रबंधक

Vimal Vadilal Shah - 087028 (DY. GENERAL MANAGER ,FQA)
(VVSHAH@NTPC.CO.IN, 9427104260) ,Simhadri

On:13/06/2023(12:17:05) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

The proposal for revision of our station LMI on “Quality management in Maintenance & Overhauling” (LMI/OGN/OPS/SYST/018) in pursuant to the revision of Operation Guidance Note (OGN) on "Quality Management in Maintenance & Overhaul" by COS vide COS-ISO-00-OGN/OPS/SYST/018 in April 2023 has been checked from FQA side.

Observations are recorded below for further examination and kind consideration.

Revised OGN Sl No	Proposed LMI Sl No	Description of Equipment for quality check	Observation
4.25.26	6.27.26	Electrohydraulic Governing System	From OGN Sl no 4.25.26.1 to 4.25.26.9 (9 nos) equipment for quality check may kindly be reviewed and incorporated in LMI (as applicable)
4.25.31.4	6.27.31.(iv)	Check for battery conditions like voltage, capacity, connecting leads, cable joints, etc.	In OGN it is considered to be (A+B) type check, in LMI it is B type.
4.25.31.7	6.27.31.(vii)	Check for changeovers in 24V power supply	
4.25.32	NA	Checking of Grounding of C&I system	In OGN it is considered, Applicability in LMI may kindly be examined
4.25.33		Checking Power Supply Monitoring	
4.25.34		Checking DDCIMS	
4.25.35		Checking UPS System	

4.25.36	Checking Chargers
4.25.37	Checking HPBP/ LPBP
4.25.38	Checking TSI System / Aux Drives
4.25.39	Checking LLD
4.25.40	Checking Servers / EWS / OWS / LVS
4.25.41	Checking Scanner
4.25.42	Checking Igniters
4.25.43	Checking Pneumatic Actuators
4.25.44	Checking Field instruments
4.25.45	Checking Analyzers (SWAS, CEMS, O2, etc.)
4.25.46	Checking Acromat / MTM / Water Wall temperatures
4.25.47	Checking Blade Pitch actuator
4.25.48	Checking APH
4.25.49	Checking ASLD
4.25.50	Checking LVS
4.25.51	Checking AHP

4.25.52		Checking Junction box / LIE / LIR	
4.25.53		Checking Aerofoil / Impulse piping for Air & Flue Gas measurements	
4.25.54		Checking Compressor	
4.25.55	NA	Checking Servers / EWS / OWS / LVS	This check is repeating in OGN at Sl. No 4.25.40. Inclusion in LMI may be examined
4.17.2.18		Gasket Replacement of gaskets except tank rim gasket as per guidelines.	After sl. No. 6.14.2 xvi mentioned in LMI next check point as per OGN seems not considered
4.17.2.25	6.14.2	Oil filling / oil after filling Check that radiator valves are open, check conservator level BDV, Moisture, content, Gas content.	Type of quality check may kindly re-consider.
4.17.2.30		Joint alarm and protection checking with operation	In OGN it is included. Possibility of inclusion in LMI may be examined.

Submitted please.

Submitted by:

आशित रॉय चौधुरी - वरिष्ठ प्रबंधक

ASHIT ROY CHOWDHURY - 101834 (SENIOR MANAGER ,FQA)

(AROYCHOWDHURY@NTPC.CO.IN, 9435027019) ,Simhadri

On:15/06/2023(12:53:01) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Pl review FQA comments.

Submitted by:

विमल वाडिलाल शाह - उप महाप्रबंधक

Vimal Vadilal Shah - 087028 (DY. GENERAL MANAGER ,FQA)

(VVSHAH@NTPC.CO.IN, 9427104260) ,Simhadri

On:15/06/2023(15:26:44) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

please incorporate the above corrections

Submitted by:

पेड्डासिंगू नाग सुरेश - उप महाप्रबंधक

Peddasingu Naga Suresh - 008111 (DY. GENERAL MANAGER ,MTP)

(PNSURESH@NTPC.CO.IN, 8332970695) ,Simhadri

On:20/06/2023(11:23:19) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

LMI Corrected and revised as per FQA comments (Annex-1 - "LMI ON QUALITY MANAGEMENT in Maintenance and Overhauling REV-2 REVIEWED.pdf"). May please process further.

Submitted by:

चौ. श्रीवेकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP)
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On:03/07/2023(11:48:06) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

After detailed examination of the LMI on " Quality management in maintenance and overhauling" in pursuant to the revised OGN released by COS, following observations are to be re-examined before finalizing the LMI.

Provided OGN Details	Considered in LMI
In OGN 4.1.1.10 Purity of Argon gas should be ensured as per IS 5760 1998 (Purity 99.996%).	In LMI 6.1.1 X) Purity of Argon gas should be ensured
Bowl Mill, Float, as per OGN class of check A.	LMI Section 6.3.6 During Assembling, ii) Worm shaft assembly, e) Float Class of check B,
In OGN 4.1.2.11 iii) Valve overhauling-lapping and blue matching above 40 NB (Class of check A+B)	In LMI to be reviewed
In OGN 4.8.7 Classifier assembly setting class of check A.	In LMI, Same item 6.3.7 XiV a) class of check B type.
In OGN 4.8.7 Roller to bull ring segments clearances Class of check A type.	In LMI to be reviewed.
In OGN 4.10.5 PA Fan Impeller welding DPT class of check A type.	In LMI to be reviewed.
In OGN 4.11.7 Impeller blade clearances (Feeler gauge) Class of check A type.	In LMI 6.5.7 Bearing+Impeller blade clearances (feeler gauge) class of check B type.
In OGN 4.13.2.4 Ball seat clearance/ Bearing Yoke clearance for self aligning journal bearing class of check A type.	In LMI to be reviewed.
In OGN 4.17.2.18 Gasket replacement of	In LMI to be reviewed.

gaskets except tank rim gasket as per guidelines, class of check B type.	
In OGN 4.20.5.4 Resistance of closing coil (circuit breaker), class of check A type.	In LMI 6.17.6 iv) Resistance of closing coil (circuit breaker), class of check B type (Random by FQA).
In OGN 4.20.9.4 Test all relays on the cubicle (Attach relevant test records), Class of check A type.	In LMI 6.17.8 Vi) Test all relays on the cubicle (Attach relevant test records), Class of check B type (Random by FQA).
In OGN 4.22.1 Random thickness measurement of the pipeline by Ultrasonic D meter. In case of reduction of thickness beyond limit on one side, rotate all such pipes by 90 deg, Class of check A type.	In LMI 6.22.1 this check is kept in B class check
In OGN 4.24.1.4 class of check is kept under A type.	In LMI to be reviewed.
In OGN 4.24.2.2 class of check is kept under B type.	In LMI to be reviewed.
In OGN 4.25.3.1 is kept in B class check.	In LMI 6.27.3 point i) & iii) only repeated twice, to be reviewed.
	6.27.54 Compressor i) P&I checking, class of check not mentioned

Submitted by:

आशित रॉय चौधुरी - वरिष्ठ प्रबंधक

ASHIT ROY CHOWDHURY - 101834 (SENIOR MANAGER ,FQA)

(AROYCHOWDHURY@NTPC.CO.IN, 9435027019) ,Simhadri

On:04/07/2023(02:24:32) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Reviewed as per FQA notings (Annex-1: 'Revised LMI ON QUALITY MANAGEMENT in Maintenance and Overhauling REV-2'), submitted for further review.

Submitted by:

चौ. श्रीवेंकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP)

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On:04/07/2023(15:35:00) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

The revised LMI (Revised LMI ON QUALITY MANAGEMENT in Maintenance and Overhauling REV-2.pdf) has been reviewed from FQA side and seems to be in order.

Submitted for further processing.

Submitted by:

आशित रॉय चौधुरी - वरिष्ठ प्रबंधक

ASHIT ROY CHOWDHURY - 101834 (SENIOR MANAGER ,FQA)

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On:04/07/2023(22:00:11) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

Reviewed and forwarded.

Submitted by:

विमल वाडिलाल शाह - उप महाप्रबंधक

Vimal Vadilal Shah - 087028 (DY. GENERAL MANAGER ,FQA)
(VVSHAH@NTPC.CO.IN, 9427104260) ,Simhadri

On:05/07/2023(09:14:02) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

अर्जुन प्रदीप नायर - अपर महाप्रबंधक

Arjun Pradeep Nair - 004419 (ADDL. GENERAL MANAGER ,TECH SERVICES)
(APNAIR@NTPC.CO.IN, 9403964693) ,Simhadri

On:06/07/2023(10:08:08) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Reviewed and submitted for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:06/07/2023(17:18:09) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Review

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:07/07/2023(11:07:45) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

LMI reviewed. Forwarded for further approval.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(RRSATPATHY@NTPC.CO.IN, 9437021584) ,Simhadri

On:13/07/2023(09:54:45) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:13/07/2023(17:13:06) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Review

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
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On:13/07/2023(18:16:36) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Reviewed from EMD side.

Inline with OGN/18 rev April 2023.

For further approvals

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:15/07/2023(18:04:12) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Reviewed and submitted for further approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:17/07/2023(09:39:17) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

May kindly review

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:17/07/2023(17:58:34) **Ref:**SMTTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Sl no. 6.18 and 6.20 may be reviewed as per Uniform Maintenance work Office Order. Office Order attached for reference.

Sl no. 6.19.5 , page - 46 of LMI is in BMD's scope.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

TANMOY SINHA - 101612 (SENIOR MANAGER ,ASH HANDLING)

(TANMOYSINHA@NTPC.CO.IN, 9434737629) ,Simhadri

On:21/07/2023(09:54:46) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

May kindly modify as commented in prepage.

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)

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On:21/07/2023(17:32:27) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Please review.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)

(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:22/07/2023(09:12:27) **Ref:**SMTTP/Maint. PLNG./2023-24/1035/MTP/203426

LMI Revised (Annx- "Latest Revised LMI ON QUALITY MANAGEMENT in Maintenance and Overhauling REV-2.pdf") as per AHP comments, but there is no change in 6.20.

Submitted by:

चौ. श्रीवेकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP)

(CHSRIVENKATAPHANIKUMAR@NTPC.CO.IN, 9440100534) ,Simhadri

On:22/07/2023(16:44:23) **Ref:**SMTTP/Maint. PLNG./2023-24/1035/MTP/203426

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)

(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:22/07/2023(17:14:53) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Sl No-6.33 for Offsite Mntc is reviewed and accepted.

Submitted for further approval.

Submitted by:

ASIS KUMAR MUKHOPADHYAY - 004586 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)

(AKMUKHOPADHYAY@NTPC.CO.IN, 9434038935) ,Simhadri

On:24/07/2023(11:20:27) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)

(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:27/07/2023(10:38:34) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)

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On:27/07/2023(10:54:05) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Pl review

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)

(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:27/07/2023(11:06:20) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

LMI Reviewed with respect to Chemistry points. Following change may please be incorporated:

Page 10 S.No. 6.1.2.10 Boiler- Inspection of drum and its internal. In Remarks column, along with Chemistry, BMD & Re-commissioning Team may be mentioned.

Submitted by:

दीपशिखा वर्मा - उप महाप्रबंधक (रसायन)

Deepshikha Verma - 007773 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(DEEPSHIKHAVERMA@NTPC.CO.IN, 9650995244) ,Simhadri

On:31/07/2023(12:05:29) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
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On:31/07/2023(16:54:06) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

6.1.2.10 - Class of check, as per OGN only B-BMD is there. In LMI Chemistry is included.
Recommissioning Team also included.

Submitted by:

चौ. श्रीवेंकट फणी कुमार - वरिष्ठ प्रबंधक

SRIVENKATA PHANI KUMAR CH - 100247 (SENIOR MANAGER ,MTP)
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On:31/07/2023(17:27:21) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri
On:31/07/2023(18:07:17) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

reviewed and forwarded.

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL HAND.- OPN)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:02/08/2023(09:51:12) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

Please review the revised LMI from O&M Civil side.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:05/08/2023(09:01:32) **Ref:**SMTTP/MAINT. PLNG./2023-24/1035/MTP/203426

Submitted by:

महेश अलुगोलु - अपर महाप्रबंधक

Mahesh Alugolu - 006936 (ADDL. GENERAL MANAGER ,CIVIL MAINT.)
(AMAHESH@NTPC.CO.IN, 9650993435) ,Simhadri

On:08/08/2023(18:14:29) **Ref:**SMTPP/MAINT. PLNG./2023-24/1035/MTP/203426

May please approve.

Submitted by:

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SIMHADRI SUPER THERMAL POWER STATION
Maintenance Planning Department

Sub: - Approval for revision-2 of LMI /OGN/OPS/SYST/018: “Quality Management in Maintenance & Overhauling” - Reg

Operation Guidance Note (OGN) on "Quality Management in Maintenance & Overhaul" was revised by COS vide COS-ISO-00-OGN/OPS/SYST/018 in April 2023. Based on this our station LMI on “Quality management in Maintenance & Overhauling” (LMI/OGN/OPS/SYST/018) is revised.

Major changes incorporated in this LMI were given in the below table.

LMI	Revised Sections (Major Changes)
LMI/OGN/OPS/SYST/018 Quality Management in Maintenance & Overhaul	1. Annexure-A: Unit overhauling norms for NTPC thermal units. 2. Annexure-B: Rolling plan for capital overhauling of Steam Turbine (200 MW / 500 MW Units) 3. Annexure-C: Rolling plan for OH/servicing of TG & Aux. equipment's during OH 4. Annexure-D: Revised Rolling plan for OH of Generators for Coal and Hydro (Koldam) 5. Annexure-E: Record sheet for trial run of motor. 6. Entire C&I QAP for OH in clause 6.27 has been incorporated. 7. Additional boiler checks incorporated in clause 6.1.2 as per OGN. 8. Necessary changes to Electrical system have been incorporated in clause 6.13 9. Checklist of Offsites areas added at clause 6.33

Put up for approval by Competent Authority

NTPC

SIMHADRI

LOCATION MANAGEMENT
INSTRUCTION

LMI/OGN/OPS/SYST/018

REVISION No- 2 DATED: June' 2023

QUALITY MANAGEMENT
IN
MAINTENANCE & OVERHAULING

Approved for Implementation: _____
HOP Simhadri

LMI REVISION APPROVAL DOCUMENT

1. TITLE OF LMI	Quality Management in Maintenance & Overhauling
2. REF. DOCUMENT CODE	OGN/OPS/SYST/018 of Apr 2023, Rev-2
3. LMI REVISION NO.& DATE	02 / J u n e '2023
4. LMI REVIEW DATE	June' 2023
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NTPC	LMI	SIMHADRI
TITLE: QUALITY MANAGEMENT IN MAINTENANCE & OVERHAULING		
LMI/OGN/OPS/SYST/018		
1.0 INTRODUCTION In today's competitive world, to stay leader in the field of power generation, production of Quality Power is a must. Quality Power is the power committed and supplied with maximum reliability, which can only be supplied by plants carrying out maintenance in a planned and systematic way. Earlier planning for maintenance was carried out only on the basis of preventive maintenance schedules but now it is also done on the basis of predictive maintenance using condition monitoring techniques. There are different techniques used for predictive maintenance of different types of equipments. Each technique works on the basis of a representative parameter whose behavior is plotted with time and based on trended information, need and timing of maintenance is planned. The main objective of planned and systematic maintenance including overhauling (see Annexure for overhauling norms) is to revive the unit health, performance and efficiency to its original or near original state such that:- - Availability is maximized - Downtime is reduced - Recurrence of shutdowns for same faults is avoided and cost of maintenance is minimized. This is generally achieved by replacing the components wherever possible following proper repair procedure. The above activities have to be planned meticulously since they are expected to be completed in a pre-decided time schedule. Since planned and properly executed maintenance helps in reducing the unplanned shutdowns and failures and in turn leads to increase in plant availability, it is utmost important to follow proper quality assurance / management norms during overhauling and other maintenance works. Before going into the details of quality management / assurance systems for maintenance and overhauling it is important to know what is meant by quality assurance. A well accepted definition of quality assurance is : “ All those planned and systematic actions necessary to provide adequate confidence that an item or facility will perform satisfactorily in service.” If we consider the other aspects of downtime or premature failure of any equipment due to inadequate quality assurance/management measures taken during various stages of manufacturing erection, testing and commissioning and finally operation and maintenance, the role of quality during maintenance phase becomes even more important. Thus it is clear that maintenance comes at the last which is expected to take care of all the unavoidable lapses which had taken place during the earlier phases. Hence, it is very essential and important to have a proper robust and well-defined quality management system for maintenance phase also.		

For maintenance phase quality management guidelines were issued by CC (OS) in 1996. Presently revised version of the same, incorporating the experience gathered in various areas in the intervening period is being issued. The implementation of the same will go a long way in fulfilling standardization efforts of the company and achievement of the forced outage rate to below 2.5%, without exceeding the planned outage target.

In the following pages description of the item along with the type of checks are mentioned. Class of check is also mentioned. Responsibility for the class of checks is mentioned at clause no.07 (implementation and responsibility centers).

2.0 SUPERSEDED DOCUMENTS:

LMI/OGN/OPS/SYST/018 , Rev 1 –Oct ' 2020.

RECORDS OF REVISION:

- Annexure-A: Unit overhauling norms for NTPC thermal units.
- Annexure-B: Rolling plan for capital overhauling of Steam Turbine (200 MW / 500 MW Units)
- Annexure-C: Rolling plan for OH/servicing of TG & Aux. equipment's during OH
- Annexure-D: Revised Rolling plan for OH of Generators for Coal and Hydro (Koldam)
- Annexure-E: Record sheet for trial run of motor.
- Entire C&I QAP for OH in clause 6.27 has been incorporated.
- Additional boiler checks incorporated in clause 6.1.2 as per OGN.
- Necessary changes to Electrical system have been incorporated in clause 6.13
- Checklist of Offsites areas added at clause 6.33

3.0 SCOPE

The scope of this LMI is to detail the quality management during overhauling:

- To ensure quality overhaul
- To ensure zero forced outage between OVERHAUAL to OVERHAUL
- To stress on meticulous inspections during the annual Overhaul opportunity

4.0 REFERENCE DOCUMENT

- OGN No. COS-ISO-00/OPS/SYST/018 Rev:2 April 2023
- Operation guidance note No.COS-ISO-00-OGN/OPS/SYST/018 Rev :1 Sept 2020 circulated by operation services NTPC corporate Centre.
- FQA manual system procedure QS-01-FQA-P-12

5.0 SPECIFIC INSTRUCTIONS:

- Finalized quality plans shall be made part of the Letter of Award / contract.

6.0 QUALITY ASSURANCE PLAN FOR OVERHAULS

For overhauling, works to be done for a particular generating unit/equipment, site activities shall be decided on the basis of pre-shutdown checks carried out, history of failure prone areas, manufacturer recommendations, findings of condition monitoring etc. Various checks to be exercised for individual equipment shall be finalized by concerned maintenance group in consultation with FQA group. The checks identified for a particular equipment to be taken up during overhaul shall be documented by the maintenance group concerned in association with FQA in the form of Maintenance Quality Plan giving details of reference documents and acceptance criteria to be followed during such checks.

As per FQA manual procedure QS-01-FQA-P-12, specialized nature of checks such as review of radiographs, ultrasonic testing, welding procedure and welder performance qualification, alignment, use of approved welding electrodes are categorized as “A” category.

Other checks are categorized as “B” category. Corporate Operation Services have issued an Operation Guidance note (OGN) no.COS-ISO000-OGN/OPS/SYST/018 Rev:02 APR 2023 detailing the checks and responsibilities and same is given below. These checks and responsibilities may be appropriately incorporated in the Maintenance Quality Plan. Finalization of Maintenance Quality Plan and categorization of the checks shall be approved by Head (FQA).

Such finalized Quality Plans shall be made part of the Letter of Award / contract

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.1	STEAM GENERATOR		
6.1.1	Pre-O/H Quality checks for Tools / Tackles		
6.1.1.1	All sky climbers to be checked and ensure its healthiness	A+B	
6.1.1.2	Scaffolding material to be checked for safety	B	
6.1.1.3	All hand tools	B	
6.1.1.4	Welding m/c	B	
6.1.1.5	Cranes, Handling equipment Trailors	B	
6.1.1.6	Winches: - Brakes Condition of wire rope Condition of pulleys	A A A	
6.1.1.7	All power points – Healthiness (including Temporary lighting)	B	
6.1.1.8	Competence & adequacy of Manpower engaged in O/H to be ensured	B	
6.1.1.9	Steam/air blowing of new tubes before installation.	B	
6.1.1.10	Purity of Argon gas should be ensured as per IS5760 1998 (Purity 99.996%).	B	
6.1.2	Post shutdown Quality checks		
6.1.2.1	Boiler Furnace/ Pent house, goose neck, Washing / Cleaning – Visual	B	
6.1.2.2	Boiler Tube Thickness Measurement: Platen superheater coil. Final superheater coil. Water wall platen assembly. Superheater coils.	A+B A+B A+B A+B	Percentage thinning allowed from original thickness - 30% for 1 st pass and 40% for 2 nd pass. However, once a coil is lifted replacement would be at -20%

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
	Reheater coil.	A+B	
	LTSH coil.	A+B	
	Economiser coil.	A+B	
	Furnace water wall (Front/Rear & Left / Right) including S-Panel.	A+B	
	Screen tube, Divisional panel	A+B	
6.1.2.3	Check on refractory / insulation, cassette baffles, binders and tube coil support and their clearance, check of the bends and Butt weld in soot blowing zone.	B	With support & surveillance by A
6.1.2.4	Sampling from all the high temperature zones for necessary micro-structure analysis.	A	
6.1.2.5	i) Welding Procedure and Welder Qualification Test prior to start of any pressure part welding in the boiler. Issue of welder identification card indicating his qualification criteria.	A	
	ii) Check of the joint fit up, cleanliness of the tube, including authenticity of raw material used in welding i.e. tube material welding consumables, etc.	B	Surveillance by A
	iii) 100% check of dissimilar joints (DPT), random radiography	B	-do-
	iv) Metallography and light microscopy for checking of internal condition of boiler tubing.	A	
6.1.2.6	100% of R.T. as per IBR on all newly welded joints (pipes), PW HT (wherever required as per approved procedure) Evaluation and interpretation of Radiographs, PWHT chart, performance of welders to be monitored and radiography percentage for tubes to be regulated based on performance of the welder.	A	
6.1.2.7	Check Hangers / support and reset if required. Hot & Cold condition reading to be used for reference	B	Surveillance by A
6.1.2.8	Check of soundness of all fillet welds and mechanically damaged area after grinding by DPT. (for headers in particular)	A	
6.1.2.9	i) Check physical condition of walkways, railings, Galleries, Buck stays, and PIN Fasteners tightness, etc.	B	

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
	ii) Check buck stays welds and Pins, proper split pins and their soundness. Replace wherever required.	B	
	iii) Headers – check physical condition, bulging etc., analysis of deposits, check for cleanliness DPT of hand hole welds, after every two years.	A+B	
	iv) Creep measurement of high temperature / pressure lines on predetermined location and compare with original values.	A+B	In case of deviation matter to be referred to OS/Engg.
	v) Furnace expansion to be checked and compared with designed value (before shutdown, during shutdown and after shutdown)	B	
6.1.2.10	Boiler		
	Drum Expansion Left () and Right () Front () and Rear () Downwards (-)	B	BMD, OPN
	-Inspection of drum and its internals, check physical condition of deposit, pitting, damage to surface and general cleanliness, check for chocking of lines i.e. CBD, dosing line, deposit in the lines, etc.	B	CHEMISTRY & Recommissioning Team
	Analysis of deposit	B	
	DPT on drum steam chest, dye to be cleaned properly after test to avoid corrosion due to presence of dye.	A	
	Angle of contact of drum with supporting U bolt and check for the condition of U-Bolt, in case of any doubt confirm by doing DPT of suspected area, every four years.	A+B	
	Check manhole door bolts, Gasket mounting area to be checked for surface preparation. Check drum inside for looseness and presence of any foreign material. Restrict entry other than concerned persons inside drum	B	
6.1.2.11	Check for completeness. Hydro test of pressure parts as per IBR/O&M manual.	B	
	a) Check safety valve seats for passing/damage.	A	
	b) Safety valve floating and setting as per manual	B	
	c) Valve overhauling – lapping and blue matching above 40 NB	A+B	

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.1.2.12	Inspection of Div., platen, FSH & RH tubes by GO gauge for any swelling.	B	
6.1.2.13	In all pendant coils with SS metallurgy Exfoliation checking to be done in bottom bends with exfoliation meter.	A+B	
6.1.2.14	PAUT of old weld joints, mapping to be done and records to be maintained properly. Where PAUT IS not possible due to space constraint issue is there RT to be done.	A+B	
6.1.2.15	All weld joints and attachment welds physical inspection to be done a qualified welding expert.	A+B	
6.1.2.16	Impulse, Vents, drains and soot blowing pipeline metal composition check by alloy analyser (PMI Machine/ XRF alloy analyser).	A+B	
6.1.2.17	Metal composition of all spool pieces inside pent house to be checked by alloy analyser (PMI Machine/ XRF alloy analyser).	A+B	
6.1.2.18	All Site welded SS joints to be checked for any Filler wire/Electrode mix up using exfoliation meter (Flux meter).	A+B	
6.1.2.19	The stub welding of all non-insulated headers to be checked by MPI in each Overhaul (Owing to higher risk of fatigue failure). For insulated headers the same has to be inspected in alternate overhauls.	A+B	
6.1.2.20	MPI of all stubs (complete 360°) of SCW C Header/ Bottom Ring header to be done during overhauling.	A+B	
6.1.2.21	Thermal flow test/ Thermal Scanning to be carried out in Water wall and all coils except RH to identify any full/ partial choking	B	
6.1.2.22	Inside oxide scale thickness measurement to be done near DMW joints on lower grade material both inside furnace and penthouse/outside furnace. In case of T91 & T23/T22 joint oxide layer thickness measurement to be done in T23/T22 material.	A+B	
6.1.2.23	Replacement of all innermost bends of all pendant coils in units having cyclic load operation. After replacement of innermost bend, MPI/UT/AET of innermost bends in every overhaul (from next overhaul onwards) and records to be maintained properly.	B	
6.1.2.24	Scanning of tubes with RFET/AET in Units having failure history due to inside pitting / deposits	A+B	

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.1.2.25	Inspection of header supports above pent house/inside furnace/outside furnace.	B	
6.1.2.26	Inspection of all fixed supports above pent house.	B	
6.1.2.27	All headers except which have been inspected twice earlier to be checked by cutting end cap/hand hole for any foreign material.	A+B	
6.1.2.28	Hardness checking of T91 and T92 joints and parent metal (on HAZ area) after completion of PWHT.	A+B	
6.2	BURNER ASSEMBLY		
6.2.1	Check for damages / erosion for the following in burner tilt mechanism:		
	i) Coal nozzle tip, seal plates, pin coal nozzle, pivot pin, roll pin and cotter pins, stationary pivot pins, drive lever pin and coal nozzle support plates.	B	
	ii) Clearance measurement on the above	B	
6.2.2	Leakage through manhole, peep holes, fuel pipe to be checked and attended	B	BMD
	Gun advance / retract mechanism, burner tilt open and its simulation with CR indication	B	C&I
	Status of ignitor/scanner and soot blower to be checked and corrected for deficiency	B	C&I
6.2.3	Check fitment of Victaulic coupling and its gasket.	B	
6.2.4	Check deflector plate position in the inlet bends.	B	
6.2.5	Check position and fitment of PF sample collection points and its sealing arrangements to avoid leakage.	B	
6.2.6	Physical checking of all orifices of P.C. piping for equal flow in all the corners.	A/B	
6.2.7	Hangers and supports, healthiness	B	
6.2.8	Dirty pitot tube test before S/D of the unit clean air flow test after O/H and corrections if any	B	BMD+OPN+EEMG
6.2.9	Scanner, Ignitor guide pipe inspection, distance measurement	B	BMD+C&I

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.3	BOWL MILL	A-FQA B-BMD	
6.3.1	Welder Qualification	A	
6.3.2	Fit up Edge preparation	B	
6.3.3	Tools / tackles Audit	B	
6.3.4	Prior to shut down i) Operating parameters ii) Vibration iii) Oil sample from gear box	B B B	
6.3.5	During dismantling i) Vertical shaft assembly a) Face out/run out b) Visual c) Dimensional ii) Pump Clearance iii) Mill bottom Visual iv) Coupling Visual	B B B B B B B	
6.3.6	During Assembling i) Vertical shaft assembly a) Visual b) Dimensional c) NDT d) Blue matching clearance ii) Worm shaft assembly a) Face out / run out b) Visual c) Dimensional d) Bearing clearances e) Float iii) Roller journal assembly. a) Visual b) Dimensional c) Bearing clearances d) Run out Float iv) Insulation cover plate assembly - Visual v) Lower bearing and pump housing – Visual vi) Pump assembly Clearances vii) Oil coolers Cleaning & hyd. Test	B B A B B B B A B B B B A B B B B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.3.7	During reassembly i) Vertical shaft assembly leveling with bull ring segments – Face out / run out ii) Air Guide vane a) Welding b) Clearances iii) Separator bottom liners a) Gap of liner b) Clearances iv) Separator body liner a) Welding b) Gap of liner c) Clearances v) Worm wheel/worm shaft Blue matching vi) Scraper assembly Clearances vii) Skirt Visual check viii) Split dust guard Clearances ix) Air seal housing Visual x) Bolt tightening Torque xii) Centre feed pipe Clearances xiii) R.E. Bellow condition Visual / DPT xiv) Classifier assembly a) Setting b) Gap c) Freeness xv) Coal pipe cleanliness xvi) Motor / worm shaft alignment xvii) Trial run for vibration xviii) Spring tension setting xix) Roller to bull ring segments. Clearances xx) Coal Mill reject rate, coal fineness and oil level leakages etc to be checked	B B B B B B B B B B B B A B B B A B B A B	By qualified welder With MTP
6.4	PA FAN	A-FQA B-BMD	
6.4.1	Tools & tackles audit	B	
6.4.2	Prior to shut down vibration	B	
6.4.3	Prior to dismantling a) Clearances b) Visual	B B	
6.4.4	Prior to reassembly a) Visual b) Dimensional	B B	
6.4.5	Impeller Welding i) DPT	A	
6.4.5	Coolers a) Cleaning b) Hydrotest	B B	
6.4.6	Bearing housing blue matching	A+B	
6.4.7	Blades and Housing –Visual and DPT	A+B	Rotor to be removed for 100% inspection
6.4.8	Bearing+ Impeller clearances (feeler gauge)	B	

6.4.9	Coupling alignment (dial gauge)	A	
6.4.10	a) Silencer insp& repair b) Trial run for vibration	B B	With MTP

6.5	FD FAN	A-FQA B-BMD	
S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.5.1	Tools & tackles audit	B	
6.5.2	Prior to shut down	B	
6.5.3	Prior to dismantling a) Visual b) Dimensional	B B	
6.5.4	Coolers a) Cleaning b) Hydrotest	B B	
6.5.5	Bolts tightness a) Blade fixing b) Hub fixing	B B	
6.5.6	Blades and Housing –Visual and DPT	A+B	Rotor to be removed for 100% inspection
6.5.7	Bearing clearances (feeler gauge)	B	
6.5.8	Impeller blade clearances (feeler gauge)	A+B	
6.5.9	Blade face and bottom fixing root DPT to check for surface defect	A	
6.5.10	Blade setting (feeler gauge)	B	
6.5.11	Governor setting (vernier caliper)	B	
6.5.12	Coupling alignment (dial gauge)	A	
6.5.13	Silencer inspection and repairing	B	
6.5.14	Trial run	B	With MTP
6.6	I.D.FAN	A-FQA B-BMD	
6.6.1	Tools & tackles audit	B	
6.6.2	Prior to shut down vibration	B	
6.6.3	Prior to dismantling a) Visual b) Clearances c) Shaft run out (dial gauge)	B B A	
6.6.4	Prior to reassembly (all components) a) Visual b) Dimensional c) Shaft UT at pin fixing area	B B A	
6.6.5	Impeller welding DPT	A	
6.6.6	Coolers a) Cleaning b) Hydrotest c) IGV housing pad blue matching	B B B	
6.6.7	Rotor leveling Housing leveling (Master level)	B B	
6.6.8	Bearing clearances (feeler gauge) + UT and DPT	A+B	
6.6.9	Impeller clearances (feeler gauge)	B	
6.6.10	Guide vane clearances (feeler gauge)	B	
6.6.11	Linkage mechanism a) Dimensional b) Functional	B B	
6.6.12	Trial run for vibration	B	With MTP

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.7	AIR PRE-HEATER	A-FQA B-BMD	
6.7.1	Check and clean the baskets	B	
6.7.2	Check for the erosion / damages (including welding) for the following: a) Cold and hot end heating element and supporting grill, radial seals, circumferential seals, rotor post seals b) Axial seals, guide bearings, support bearing, carbon ring nozzles and piping, metallic expansion bellow c) APH drive systems gearboxes, couplings, pins and comes, etc.	B B B	
6.7.3	Clearance measurements of the following a) Guide bearing and support bearing. b) Axial seal clearances c) Circumferential and radial seal clearances d) Drive alignment e) Rotor leveling f) Root gap and Bottom gap of drive -Monitor the condition of Trunion bolts rack pinion alignment -APH pressure differential and heat pick up across APH, parameters of primary and secondary air, flue gas etc and O2 level in flue gas need to be checked based on operational feed back and accordingly seal adjustment to be done.	A+B A+B A+B A+B A+B A+B B B	
6.8	GRAVIMETRIC FEEDER	A-FQA B-BMD	
6.8.1	Check for physical condition of feeder parts for damages / erosion, replace where necessary	B	
6.8.2	Check head pulley, take up pulley Tension roll, Belt span rollers, weight rollers, conveyor chain (for gravimetric feeders). Check the drive chain and sprocket tightness, teeth health, gear box, bearing condition, Healthiness of feeder I/L& O/L gates	B	

6.9	POWER CYCLE PIPING	A-FQA B-BMD	
6.9.1	Check for hangers / support for preset values in cold and hot condition	B	
6.9.2	UT of MS/HRH/CRH weld joints, sampling points, weld joints randomly	A	
6.9.3	Blue matching of MS line valve seats and DPT for any surface defect for identified valves as per requirement	A+B	
6.9.4	Microstructure replicas of power cycle piping and headers	B	
6.9.5	Thickness survey on bends of power cycle pipings, feed water lines -Random DPT on Bends	A A	
S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.10	TURBINE	A-FQA B-TMD	
6.10.1	EOT Crane a) Crane maintenance before shutdown as per requirement b) Testing using dummy load c) Deflection, Condition of wire rope sling, Hook opening, Wire rope Drum etc	A+B A+B B	
6.10.2	Before shutdown a) Vibration measurement on all bearings at full load and transient data during coast down & temperature of all bearings b) Review of operating parameters and cylinder efficiency tests c) Testing of TG Bearing lubricating oil d) Governing characteristics e) Insulation survey and thermal mapping of Power cycle and TG integral piping	B B B B B	With MTP OPN & O&E CHEM
6.10.3	Before & during dismantling a) Alignment of all the TG Rotors & MOP b) Oil clearances of all the bearings c) Bump check, roll check, horn drop tests and swing check of HP front rotor and exciter rear rotor. Initial slope measurement at all the rotor journals of TG, bump check, roll check, horn drop tests and swing check of HP rotor d) Check for Cold / Hot setting values of all critical hangers and necessary adjustment alignment of piping with	A+B A+B A+B B A+B	

casing.		
e) Correction for Catenary deviation at deck spring elements to achieve desired catenary	A+B	
f) Rotor system catenary checking at all pedestals / rotors	A+B	
g) Welder & procedure qualification test prior to any welding	A	
h) Measurement of Bridge gauge readings at bearings and its comparison with erection /last overhaul values.	A+B	
i) Dismantling steam flow path measurement/audit	B	
j) TG deck Condenser spring deck survey.	B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.10.4	Rotors i) Check for all the rotors run out at journal, parting planes, surface roughness, pitting and dents (Electrical), pitting and dents (Corrosion) ii) Coupling face run out, coupling face flatness iii) Knock and sound test for intactness of stellite strip erosion shield, lacing wire iii) looseness of blade with shroud and rigidity, gaps between shrouds. iv) Visual /DP test for surface defect and intactness of brazing of lacing wire, porosity in brazing, for loose lacing wires / lacing wire roundness v) Condition of stellite strip i.e. erosion Status - repair / replacement required, in case of replacement of stellite strip, brazing and soldering to be done as per approved procedure vi) Steam flow path clearance of rotors. vii) Alumina blasting of rotors, surface roughness before and after sand blasting In situ PAUT or MPI/NFT of LP last stage blade as per Scope of work.	B B A+B A B A+B B A	
6.10.5	Bearing & Journal i) Measurement of slope (catenary of rotor) at all the pedestals ii) All dimensional checks iii) U.T.+DPT of T.G. bearings for checking of bonding between white metal and parent metal iv) Blue matching of bearing pads and bedding of the bearings with rotors. Bearing support pieces to pedestal and support plate with TOROUS v) Blue matching of Bearing Side pads, keys & Packers vi) Ball seat clearance / bearing yoke clearance for self aligning journal bearing	A+B A+B A A+B B A	
6.10.6	Casing & Diaphragm i) Visual and NDT inspection including MPI of all rotor blades, shrouds, casing, strainers, MOP shaft and impeller, bearings etc. Rectification of defects and general cleaning. ii) All dimensional checks including parting plane gap of HP & IP cylinder	A B	

6.10.8	Final check and general precaution before boxing up i) Final alignment of turbine rotor & MOP& final catenary check ii) Bump ,Roll, Horn drop check and axial float of rotors. Feeler check at HPT front & IPT rear for see through seals & casing centring iii) Side clearance, top oil clearance of rotor journals at all bearings iv) Bridge gap readings at all the bearing journals after overhauls v) coupling bolt stretching vi) Final coupled run out of IP/HP, IP/LP,LP/Gen/exciter(journals, couplings/ IP neck &LP neck) . HP front &Exciter swing check vii) Reaming & Honing of coupling holes, coupling bolt clearances& weight balance viii) Microstructure replica to be taken on TG rotor, HP/IP cylinders, nozzle box, liners etc for turbine which has completed 1,00,000 hrs of operation ix) Check for proper cleaning and absence of any foreign material before boxing up of turbine JOP shaft lift, JOP header pressure, Individual Bearing JOP pressure	A+B A+B A+B A+B A+B B B+A A+B B A A A+B	
6.10.9	Other important checks i) Checking of Pedestal foundation for cracks, voids etc. (in case of spring deck foundation all checks to be done as per check list) ii) Chemical analysis of deposits on turbine blades iii) Steam strainers inspection, Measurement of DP before and after overhauling iv) Cleaning of MOT gauge glass etc. and attending the defects v) Representative impression of the uniformly eroded blades of the turbine stage. For example wet steam erosion on leading edge of blade vi) Clean and inspect lub oil coolers for damage. Hydrotest and plug leaking tubes vii) Freeness of pedestal movement. Check pedestal keys fits & tolerances	B B B B B B B	

6.11	TURBINE AUXILIARIES	A-FQA B-TMD	
6.11.1	i) NDT of all the Rotors of Booster Pump, BFPs, CEPs, ACW Pumps and PSW pumps ii) Hydraulic test of coolers & cooling jackets iii) Run out of individual Rotors of Booster Pumps, CEP, ACW&PSW and their final alignment with motor iv) Internal clearance measurement of all the pumps viz BFPs, CEPs, Booster Pumps, ACW Pumps, etc. v) Blue matching, seating of valves & NRVs vi) Water fill test of tube nest of LP & HP heaters vii) Tube cleaning of condenser viii) Flood test of condenser	A A A B B A B B	 With OPRN With OPRN
6.12	BOILER FEED PUMP (BOOSTER / MAIN)	A-FQA B-TMD	
6.12.1	i) Decouple the booster pump and dismantle for servicing ii) Clean and check the internals i.e. impeller, wear rings, shaft, sleeve, etc. change the worn out parts iii) Clean and check the bearing condition DP test of impeller wear ring surface area of contact (blue match), replace / repair iv) Clean the bearing cooling chamber inlet and outlet CW piping -Check mechanical seal faces for damage lap / replace the seal and check the mechanical seal, gasket for leakage. Check seal compression v) Box up the booster pump maintaining specified clearances. Replace parting plane gasket vi) Drain the hydraulic coupling, clean and check its parts for wear. Put specified grade of fresh lubricant as per laid down norms. Check gears for backlash check brgs (blue match) vii) Align booster pump with motor viii) Decouple the main pump and dismantle for servicing ix) Remove the main pump cartridge. Clean and inspect for damage / wear. Replace if required x) Clean and check bearings for damage. Replace / colour match bearings and adjust the specified clearance as per	B A+B B A A+B A+B B A+B B B A+B	

	requirement xi) Check the mechanical seal of the main pump, lap / replace the seal the test the mechanical seal gaskets for leakage xii) Clean and check the seal coolers for tube conditions. Hydro test and plug the leaky tubes. Change the gaskets / O ring as required xiii) Box up the main pump maintaining the specified clearance xiv) Align the main pump with motor xv) Service / Replace valves / orifices on CW line / oil line as required xvi) Clean and inspect lube oil & working oil coolers for damage / checking of tubes. Hydro test and plug leaky tubes xvii) Clean lube oil tank, gauge glass, etc. and attend to the defects xviii) Trial run of the pump with drive	A+B B A+B A A+B A+B B B	
6.13	GENERATOR	A-FQA B-EMD	
6.13.1	Before shutdown I. Checking of all the minimum critical Spares, Consumables better with healthiness (Rubber cords & O-Ring with good shelf life) & Service required for carrying out the Overhauling. II. Capturing of all liabilities in Generator & its Auxiliaries. Leakages attended if any on temporary basis. III. Record of all the Parameters of Generator & Auxiliaries from CCR & Local. Protocol to be made & deviation shall be recorded for attending during overhauling (H2 Consumption, Purity, Defective Instruments, Parameter out of design range Etc, coolers in service). IV. Checking of CO2 evaporator functioning	B B B B	Clean condition to be ensured as per OGN/OPS / ELEC/020 OPN OPN
6.13.2	During shutdown Barring gear checks I. H2 leak areas (Bushing, CC Bolt, End shield, Gas Pipelines joints etc) which are not planned for opening II. Purge out of H2 From Generator. III. Check slip ring smoothness & dimensions (Brush gear / Rotor E/F), necessary polishing / cutting to be taken up before barring gear stop if Value (Run out Shall be <0.05mm)	B B A	

6.13.3	Dismantling Checks I. Recording of all the initial clearance during disassembly of generator Like CRO, Swing check, alignment reading, bearing clearances (SOC, TOC, top clearance with saddle) Seal bore at outer oil catcher region, axial readings of rotor face to end shield, diode wheel bottom cover, compressor/fan radial & axial reading, magnetic center. II. Check coupling faces and journal surfaces for uneven wear, rubbing, discoloration, spark erosion, electrical pitting marks III. Check condition for oil/dirt contamination, particle deposits insulation and or metallic debris from abrasion fretting, foreign material hit marks, discolorations from overheating and signs of electrical arcing cracks IV. Photography of all the initial observation.	B B B B	
6.13.4 6.13.4.1	Rotor Thread out Stator a. Healthiness of generator stator temperature points & other instruments points b. Check for dust accumulation, greasing, damages, de colorization or mechanical damage on core teeth, Lamination, Core Packets, End winding bar & ring bar, PFTE Hose & bore area. c. Check for crack & relative movement of core suspension assembly, end windings and ring bus bar. d. Check for missing spacers block, lashings and rope ties in end winding area & its support system. e. Check for signs of partial discharge gray/white powder in the locations of line end coils to the neutral coils , slot exits of bars, connecting bus-bar lacings with support, inter-phase support packing's. f. Glue injection on overhang winding in 200MW machine g. Inspection of air-gap seal/baffle ring for any crack/damage. h. Replacement of cords/O-rings, with new set. i. Lock/spring washers are to be used in bolt head/nuts of Gas cooler water box (inside generator). j. During seal ring machining, the diametrical difference (between ring ID and rotor shaft OD) shall be kept minimum range	B B B B B B B B B A	C&I

	of designed value. Ring ID measurements in (by Inside micrometer) shall be calibrated by outside micrometer which is used for measurement shaft OD k. Measurement of Oil catchers, Seal body, seal ring, bearing & Blue matching.	A	
	Borescope Inspection a. Ensuring healthiness of ventilation spacers in core packets & blockage of the core cooling ducts. b. Inspect for core burning and a white, light brown, or yellow powder along the slot wedges. c. Inspecting coil side for damage of semi conductive coating. d. Presence of foreign materials in narrow spaced in between end winding e. Inspection of bushing (hydrogen cooled) inner bore for oil accumulation, blockage by grease or foreign materials and crack. f. Ensuring healthiness of ventilation spacers in core packets & blockage of the core cooling ducts. g. Inspect for core burning and a white, light brown, or yellow powder along the slot wedges. h. Inspecting coil side for damage of semi conductive coating. i. Presence of foreign materials in narrow spaced in between end winding j. Inspection of bushing (hydrogen cooled) inner bore for oil accumulation, blockage by grease or foreign materials and crack.	B B B B B B B B B B	
6.13.4.2	Rotor Visual Inspection a. Check rotor barrel surface for any oil/dirt contamination, particle deposits, insulation / metallic debris from abrasion fretting, foreign material hit marks, differences in surface colors or shades, tooth or wedge discolorations from overheating, Signs of electrical arcing at wedge ends, over heating at cut ends on pole cross-cut due to negative sequence currents b. Check teeth surfaces and wedges for transverse cracks, cracks at the rotor bottom, coupling faces and journal surfaces for	B B	

	uneven wear, rubbing, discoloration, spark erosion, electrical pitting marks, Coupling bolt holes for wear, deformation. c. Check fan blade attachments for signs of looseness, fretting wear, overheating, shrunk on fan hubs for signs of fretting and cracking, balance weights for signs of movement (looseness), staked and locked d. During disassembly of the CC-bolt, the condition of the forging bore thread to be checked for any damage. Required cleaning of the thread to be carried out with matching tap. During assembly, the locking nut to be tightened with proper torque specified by OEM. Sufficient curing time for resign must be given. This is applicable for BHEL-500-THDF/210-THW/210-THRI generators. e. Signs of Retaining ring movement in service, axial or circumferential, inspection for corrosion cracks & removal of retaining ring as per requirement / Recommendation f. Polishing of shaft surface near oil catchers, Seal Assembly & bearing	B B B B	
	Borescope Inspection a. Check for surface deposits of oil, insulation or metallic dust on shaft, balance ring and on winding coils. Analysis of deposits to be carried out for their origin (core abrasion, rubbing of aluminum gas baffles, rotor copper dusting, etc.) and addressed. b. Check for coil radial alignment, any slippage between turns, tilted coil stacks, copper distortion from creep growth. Signs of distortion of bottom cross-over pieces, cracking of copper, Signs of inter-turn insulation slippage. c. Check for end winding blocking, movement from original placing, cracked blocks, obstruction to ventilation passages, any foreign object d. Check the condition of lead connection wedges from the studs to "goose neck" lead under the RR. Check for lifting of the last wedge from its slot. Look for signs of fretting abrasion deposits along wedge-slot interface e. Wedge axial migration, signs of blockage of gas ventilation holes; if feasible, with optic probes check turn insulation condition at vent holes; look for turn insulation migration, breakage, overheating	B B B B B	

	f. Inspection of J-strap for connection CC-bolt.	B	
6.13.4.3	Exciter a. Check contamination of the Exciter unit for dust, conducting particles and abrasive particles, Mechanical damage due to foreign material, broken band tapes, tape delaminating or signs of broken/cracked epoxy bonds, cooling ducts blocking and ventilation restriction, wear out mark. b. The flexible lead connections from the rotor winding to the diode wheel are supported against the centrifugal loading. They should be checked for signs of looseness, crack mark, distortion and insulation abrasion from vibrations. c. Stationary field coils (poles) are to be inspected for mechanical damage, insulation peel off, overheating. d. The wheel inspection includes checks for the diode assemblies' component looseness and contamination with dust and oil. All connections should be examined for tightness. All fuses should be checked for their aging in addition to checks for blown fuses. All diodes and heat sinks should be checked for mechanical damage and chipping. checked for any foreign object inside diode wheel. e. Check coupling face and journal surfaces for uneven wear, rubbing, discoloration, spark erosion, electrical pitting marks. multi contact tape and pin for any sign of overheating and deep groove/cut/rubbing mark on it. Contact pin surface shall be free of deep cut/roughness. Same inspection is applicable machines having current carrying wedges. f. Check the locking key of exciter blower (if not shrunk fit). DP test is to be done on blower sitting area. g. Inspection of main exciter neutral point	B B B B B B B	

	connecting cable to winding side and slip ring side for integrity of brazing point, looseness of connection etc. h. Inspect the condition of the slip rings, Rotor E/F sensing brushes, brush holders and brush holder brackets. Check condition multi contact tape. i..Inspection of cable from brush holder bracket to junction box, for any cut/damage. j.Checking of inter-pole electrical connections any crack /looseness. k. Check PMG stator for contamination with oil and dirt, causing overheating and aging of the insulation. Overheating in the end winding will be noted by discolored, cracked and embroiled insulation. l. The end winding of PMG stator should be inspected for signs of loose wires, vibration, and wear damage m. Check any rubbing mark on the PMG stator n. Check condition of fuses for heating marks, may be replaced in 12 years as precautionary measure. o. All magnets and their supports should be inspected for signs of developing looseness such as fretting debris, signs of fatigue crack and tightness of all mounting bolts. p. During disassembly of the CC-bolt, the condition of the forging bore thread to be checked for any damage. Required cleaning of the thread to be carried out with matching tap. During assembly, the locking nut to be tightened with proper torque specified by OEM. Sufficient curing time for resign must be given. q. Balance weights to be properly locked.	B B B B B B B B B B B	
6.13.4.4	Auxiliary systems a. Before dismantling of oil float valve proper identification & marking to be done. b. Cleaning of seal oil, Stator Water cooler /PHE and Hydraulic test and Filter. c. Stand By Cooler / Filter to be taken into service after proper flushing. d. Re-paint of pipeline joints after DP & replacement of pipe line Gaskets with new gaskets	B B B B	

	e. Pipeline Hanger support to be Checked & as per need to be adjusted. f. Cleaning of seal oil tank/ vacuum tank/intermediate tank/seal oil storage tank. (as per applicable) g. Inspection of impulse line rubbing with other pipeline/grating/ metallic structure & flushing to be carried out to remove any blockage. Care to be taken during connection of impulse lines. h. Calibration of Pressure gauge/transmitters/switch, level transmitter and flow meters (as available). i. Maintenance of critical valves Pressure Relief, Pressure reducer valve, DPR, Equalizing valve etc. as per requirement. j. Cleaning and servicing of Hydrogen gas driers	B B B B B B	Check for source of leakage and rectify. Any welding to be done by lowering the oil level and should be done by qualified welder using approved welding procedure.
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6.13.5 Testing

	Electrical test	Mechanical other test	Checks
Generator stator	● IR value ● Winding resistance ● ELCID ● Wedge deflection test ● Tan delta ● HV Test as per Recommendation	● Visual inspection ● Hydraulic test of stator windings and PW circuits ● Seal oil inlet pipeline in the end shield ● DPT of Air Gap Seal Fixture ● NTF stator Overhang (Except BHEL 500MW) ● Torque tightness of stator top & bottom Bar joint, Bushing bolts HEL Make (500,660,800Mw) ● Generator ATT	A
Compressor /Fan	● Compressor segment IR Value	● Fluorescent DPT of compressor blades moving & Fixed blades ● DPT of End shield mounted fan guard	A
Generator Rotor	● IR value ● Winding resistance ● AC Impedance ● Digital RSO ● Voltage balance	● Visual inspection ● Fluorescent DPT of retaining rings ● UT of retaining rings ● Gas leakage test of CC bolts ● Ventilation duct Purge Test / possibility test in 200MW BHEL / Electrosila	A
PMG	● IR value ● Winding resistance	● Visual inspection	A
Rectifier Wheel	● IR value ● Fuse resistance with Micro Ohm Meter ● Reverse leakage current test	● Visual inspection ● Gas leakage test of CC bolts	A
Main Exciter Field	● IR value ● Winding resistance ● AC Impedance ● Pole drop test	● Visual inspection ● DTP OF exciter Fan blower	A
Main Exciter Armature	● IR value ● Winding resistance	● Visual inspection	A
Bearing pedestal	● IR value of insulation	● DPT ● Side Pad Blue Matching	A
Coolers		● Exciter-Hydraulic ● H2 Coolers-Hydraulic ● Seal oil Cooler-Hydraulic ● Primary water Cooler-Hydraulic	A
Seal Ring		● DPT ● UT ● Dimensional Checks	A
Seal oil		● DPT Float	A
Pipelines		● DPT of All welding Joints	A
Bearings		● DPT	A
Stator & Rotor	Daily trending of IR value		B (Maint.)
Any other test	As per OEM / COS Recommendation	● As per OEM / COS Recommendation	A

SL. NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHEC K	REMARKS
6.13.6	Assembly a. Generator Rotor Threading in Protocol to be made with completion of all the test & results. Record of IR value. b. Stator & Rotor thorough cleaning and inspection is carried out without any abnormality. c. Generator stator & Rotor air blowing, vacuum cleaning & Painting / Varnishing carried out before rotor thread in. d. Stator bore to be checked from both TE & EE for any foreign before mounting air gap seal/baffle rings e. During assemble stage wise clearance to be measured & logged in their respective formats before taking up the next activity along with no foreign object left in confined spaces to be checked & recorded. f. The condition of sealing rubber cords/gaskets (used for end shields, seal body and gas coolers) shall be checked before usage. Use of rubber cords/gaskets preserved more than three years (even if kept in AC room) shall be avoided. g. Clearance of fan with guard and compressor assembly. Set of reading of blade tip to stationary part shall be taken in four position (0-90-180-270) of shaft. h. IR check of inner oil catcher, seal body and outer oil catcher after centering. i. Before inserting Gas coolers, physical inspection for objects shall be checked. j. Seal oil inlet line, drain lines (inside end shield, seal body) shall be cleaned and ensured that no foreign object inside pipe before assembly/ connecting to external line. All Pipe lines shall be checked with air to ensure no blockage of pipelines. k. Impulse lines connected to seal body shall properly be identified to negate any chance reverse connection (between Hydrogen & air side). l. Clearance measurement between slip ring with tip of brush holder, m. All Seal oil & Stator water parameters (temperatures and pressures with desired head adjustment) is available in healthy state in DDCMIS seal oil page.	B B B A A B B B B B B	

	n. P&I checks of Stator Water Pumps, AC seal oil pumps (main and standby) and DC seal oil pump by actual simulation to the extent possible. o. Seal oil system is to be kept OFF while doing TG bearing Lube oil flashing is being done. p. Bus Duct to Generator Bushing flexible connection bolt torque tightness q. Busduct pressurization system condition. Pressure hold timing to be recorded r. Busduct and its associated auxiliaries' inspection and testing as per OGN/OPS/ELEC/039 s. Numerical relays healthiness (manual DR trigger and checking of binary/analogue inputs, taking working backup) Refer OGN/OPS/ELEC/040 t. GRP Protection scheme and logics Refer OGN/OPS/ELEC/040 & OD-OPS-ELEC-003 1. Testing of numerical relay to verify setting and logics. Testing of all other electromechanical/static relays as per installation. Checking of all lockout relays. 2. Verify all the inputs & outputs (binary/analogue) are healthy 3. Check the final setting of relays as per master record 4. Joint checking of DDCMIS alarms along with SOE	B B B A B B B B	
6.13.7	Generator Barring Gear Operation a. Barring gear Clearance to be jointly signed containing • Assembly log sheets • LP turbine - Generator alignment and foundation bolt tightening • LP turbine - Generator coupling bolt tightening • LP turbine- Generator CRO • Exciter – Generator alignment • Exciter swing check • Assembly of Generator bearing • Boxing up Generator • Boxing up exciter and assembly of walls and covers. • Bearing & Seal oil flushing & in service • P&I Checks Completed • C&I Instruments availability ensured b. Protocol for Boxing-Up of Generator signed for closing the manholes after joint	B(EMD +TMD +C&I+ OPRN)	

	inspection for any foreign material, no abnormal rubbing sound, or no seal oil entry inside the generator after machine put on barring gears. c. Adjust all the parameters of Seal oil as per requirement. d. Air Tightness Test of Generator to be carried taking temperature & ambient Pressure variation into consideration. Generator, Auxiliary system along with the pipeline to be kept in service for ATT Proving. e. Stator water makeup to be done & necessary protocol to be jointly signed by Operation, Chemistry & EMD f. Adjust all the parameters of Seal oil as per requirement. g. Generator & Auxiliary system along with the pipeline to be kept in service for ATT Proving.	A B(EMD +OPR N) A B(EMD +OPR N+Chem) B(EMD +OPR N) B(EMD +OPR N)	
6.13.8	Rolling & Synchronization a. AVR checks (Calibration, Settings, Jumper / Switch setting to checked & recorded), Dummy load test before clearance for rolling. All The SOE, Alarm, events to be recorded with time stamping for Generator protections, Breaker (EHV, FB) Inter trips - To be jointly checked & protocol to be signed by C&I Electrical and Operation. b. Check for any mechanical rubbing sound at Soaking speed & ramp up. c. Check the PMG voltages & phase sequence d. At 3000RPM adjustment of Stator Water (winding & Bushing), seal oil parameter as per requirement. e. Dynamic test of AVR Channel Change Over, Response. GRP Protection checking	B B(EMD +C&I+ Oprn) B B B B	
6.13.9	Reports & Protocol. a. Respective HOD ensure attachment of standard Protocol & measurement log sheets, P&I Checks, Alarm, SOE Event Checks etc as attached with respect each make & model of machine in the station while putting up the LMI for approval	B(EMD)	

	(Any deviation to be recorded). b. Responsibility concerned HOD shall hand over soft copy of the protocol, Logs, P&I checks jointly signed and report of any observation and brief rectification taken up with Photo graph to MTP. c. HOD MTP shall make the softcopy available on internet for ready retrieval.	B B	
6.14	POWER TRANSFORMER	A-FQA B-EMD	
6.14.1	Before shut down i) Oil a) Review of past DGA results b) BDV moisture content, Tan delta, specific resistivity ii) Oil Temp indicator / winding temp. indicator a) Review of readings for any abnormality iii) Transformer a) Operating hourly record since past overhaul overloading period b)Silica gel colour iv)Thermography of transformer for detection of hot spots	B B B B B B	Replacement if aged Oil filtration to be decided on value of O2/N2 in DGA, BDV and moisture values
6.14.2	After shut down		

	i) Transformer		
	a) IR of different windings to earth & between different windings. All other electrical tests at low voltage to be checked	B	
	b) Magnetising current	B	
	c) Magnetic balance	B	
	d) Ratio test	B	
	e) Winding resistance test	B	
	f) Tan delta test	A+B	
	g) Cooler supply changeover scheme check	B	
	h) SFRA test signature matching & analysis	B	
	i) WTI calibration	A+B	
	j) Bushing tan delta test , tap earthing check by resistance method	A+B	
	k) Checking of drain pit to main pit	A+B	
	l) Marshalling box TB tightness check	B	
	l) BCT characteristic checking	B	
	m) Neutral connection tightness check	B	
	n) IR of all trip cables	B	
	ii) Oil		
	a) BDV & Moisture content	B	Centrifuge the oil if required
	b) Level in conservator & bushing	B	
	iii) Tank, radiator, piping valve		
	a) Check for leakages	B	Leaking points to be recorded for repair /change of gaskets

	iv) Tap changing gear a) Check for any damage and electrical / manual operation	B	Any damage / mal functioning should be recorded and repaired in consultation with the manufacturers
	v) Tank a) Check for marks of oil leakage	B	Check for source of leakage and rectify. Any welding to be done by lowering the oil level and should be done by qualified welder using approved welding procedure.
	b) Check for marks of rusting	B	Plan for painting
	vi) Bushing a) Check for any cracks / chippings	A	
	b) Oil leakage after applying N2 pressure on conservator	A	
	c) Tan Delta test	A	
	• Check metal links for any heating mark • Check earth after any spark mark • Tan delta test tap earthing to be ensured after the Tan delta test is over	B	
	vii) HV / LV Terminals a) Check for any heating marks check use of bimetallic washers	B	
	viii) Bucholtz relay a) Manual operation check	A+B	
	b) All other functional relays like PRDS, oil / winding temp. alarm and trip etc should also be checked for correct position	A+B	
	c) Replacement of Buchholz relay, PRV, OTI/WTI, SPR, OSR, as per guidelines	B	
	ix) MOG a) Check operation while lowering oil level of conservator/by simulation	B	
	x) WTI & OTI after dismantling the probe	B	

	a) Calibration check at ambient temp and at elevated temperatures, check for all the electrical protection. & BCTs xi) Oil separator rubber bellow If gauge glass level is visible, then check for any puncture by air pressurizing conservator after draining oil.	B B	
	xii) Marshalling Box a) Check for any damage / dirt b) Attend hotspots of MB components identified during pre-shutdown thermography. Replace MP components as per guideline. xiii) Cooling system, Oil Flow Indicator a) Operation of oil pump and fan to be checked xiv) Valves a) Check for any seepage when closed xv) Oil after transfer to storage tank BDV, Moisture content, IFT, Tan d, colour, acidity sludge content xvi) Transformer internals Check for physical condition / foreign bodies as per guidelines xvii) Gasket Replacement of gaskets except tank rim gasket as per guidelines xviii) Core clamp IR at 1 KV xix) Radiators and pipeline Check for leakage marks and to be attended xx) Pump with motor fans IR of winding test xxi) Tank Filling and monitoring of N2 pressure for 24 hrs if transformer is to be kept without oil xxii) Transformer after hot air circulation and vacuum for 24 hrs Dew point measurement xxiii) General assembly a) Check correctness of HV/LV link connection, bolt tightness using torque wrench. b) Clean the oil pit and drain line to main pit. xxiv) Oil filling / oil after filling Check that radiator valves are open, check conservator level BDV, Moisture, content, Gas content. xxv) Transformer a) IR of different windings to earth	B B B B B A+B B B B+A B B A+B B B+A A+B A B+A B	

6.15	HT MOTORS	A-FQA B-EMD	
6.15.1	Stator i) Stator Core Physical condition cleanliness, hot spot, insulation coating, absence of any foreign material ii) Stator Winding Physical Condition Cleanliness Fixing arrangement RTD iii) Stator Wedges / Slots a) Physical condition b) Looseness / tightness iv) Stator overhang winding including bracing v) Stator insulation Physical condition to take necessary action based on inspection results	A+B A+B A+B A+B A+B B B B B	
6.15.2	Rotor i) Shaft journal surface Physical condition, take corrective action in case of scoring pitting, DPT for crack	B	

	detection ii) Rotor Winding Cleanliness, hot spot, wedge looseness, rotor brazing (in case of doubt DPT of brazing) iii) Short Circuiting Rings Physical condition and cleanliness iv) Rotor insulation Physical condition and cleanliness	A+B B B	
6.15.3	Terminal Box Physical condition, cleanliness, sealing gasket and bushings connection tightening	B	
6.15.4	End Shields Physical condition, cleanliness, replace in case of crack in seals	B	
6.15.5	Stator cooler Physical condition cleanliness, hydro test to ensure leak tightness	B	
6.15.6	DE Bearing		
	Physical condition of the bearing, white metal bonding, RTD etc. cleanliness	A+B	
6.15.7	NDE Bearing		
	Physical condition, white metal bonding, condition of seals, replace if damaged, bearing insulation pad condition, RTD etc. cleanliness	A+B	
6.15.8	Space heaters		
	Healthiness, replace if needed	B	
6.15.9	Earthing strips / connections		
	Check physical condition, looseness ensure double earthing whenever applicable	A+B	
6.15.10	Foundation bolts		
	Physical condition, looseness	B	
6.15.11	Motor parts clearance		
	i) DE Bearing Radial clearance	B	
	ii) DE Bearing Housing interface	B	
	iii) NDE bearing Radial clearance	B	
	iv) NDE bearing housing interface	B	
	v) Air gap	A	
	vi) IR value (R1/R2, B1/B2, Y1/Y2, R-Y, Y-B, B-R)	A+B	
	vii) Winding resistance (R1/R2, B1/B2, Y1/Y2)	B	
	viii) Space heater	B	
	ix) Space heater resistance	B	
	x) Bearing insulation	A	
	xi) Coupling alignment before dismantling	B	

	xii)RTD resistance	B	
6.15.12	Recommissioning of checks		
	i)Rotor assembly freeness	B	
	ii)Bearing lubrication OK a)DE Bearing b)NDE bearing	B B	
	iii)Earthing connection & cable connections OK	A	
	iv)Motor cooler fitted / fan fitted check for direction of rotation	B	
	v)C.W. flow to coolers OK & all valves operable smoothly	B	
	vi)IR value checked & recorded	A	
	vii)General cleanliness around equipment OK	B	
	viii) Start the motor and a)Check direction of rotation in reference to driven equipment OK. b)No oil leakage c)C.W. Flow to bearings / coolers d)Sound level OK / abnormal e) Vibration level OK / bnormal	B B B B B	MTP
	ix)Check protection and interlocks	B	OPN
	x)Run the motor for 4 hrs. on no load and record parameters in log sheet (attached)	B	

Record sheet for trial run of motor

Time	Starting Current (Amp)	Speed (RPM) (Amp)	No load current	Bearing DE	Temp NDE	Bearing vib.		Engr.in charge	Remarks
						DE HVA	NDE HVA		

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.16	LT MOTORS (applicable for motors for which overhauling is planned)	A-FQA B-EMD	
6.16.1	Shaft / rotor		
	Check for physical condition, cleanliness insulation condition any rubbing or mechanical damage marks	B	
6.16.2	DE Bearing		
	Physical condition, grease condition	B	
6.16.3	NDE bearing		
	Physical condition, grease condition	B	
6.16.4	Stator/Windings		
	Physical condition, cleanliness, wedge slot tightness, insulation condition, mark of rubbing/damage	B	
6.16.5	Foundation bolts		
	Physical condition, tightness, any damage etc.	B	
6.16.6	Earthing strip / connections		
	Physical condition, earthing strip tightness, any surface defect	B	
6.16.7	Junction Box		
	Cleanliness, loose connection (if any)	B	
6.16.8	Motor Parts Clearance		
	i) DE Bearing radial clearance	B	
	ii) DE bearing housing interface	B	
	iii) NDE bearing Radial clearance	B	
	iv) NDE bearing housing interface	B	
	v) IR value (R/E, B/E, Y/E, R/Y, YB, BR)	B	
	vi) Winding resistance (R1/R2, B1/B2, Y1/Y2)	B	
	vii) Heater winding IR	B	
	viii) Heater winding resistance	B	
6.17	HT/LT SWITCHGEARS	A-FQA B-EMD	
	Physical condition, cleanliness, lubrication	B	
6.17.1	Circuit breaker		
	Physical condition, cleanliness, lubrication, timing contact resistance	B	
6.17.2	VT housing		
	Cleanliness	B	
6.17.3	Interlock relay panel		
	Cleanliness	B	
6.17.4	Secondary wiring & fuses		
	Physical condition, tightness of fuse	B	
	Note: Check the breaker operation on test / local / remote Trip the breaker on protection Check the final setting of relays	B	
6.17.5	Bus bar		
	Physical condition, cleanliness Bus bar tightness Torque (as per Manufacturer Recommendation)	B B+A	

	Conductivity test (Ductor test) Final IR measurement of the bus bar	A B	
6.17.6	Circuit Breaker		
	i) Correct entry	B	
	ii) Slow closing mechanism	B	
	iii) Racking arrangement	B	
	iv) Resistance of closing coil	A	Random by FQA
	v) Trip free & Anti pumping operation	B	
	vi) Grease mech. Parts	B	
	vii) check Mech. Key interlock	B	
	All excess openings are blanked	B	
	Panel identification levels provided on both front & back	B	
	Checking of interrupting chamber healthiness	B	
	Checking of contact wear of the moving contacts by means of contacts wear indicator	B	
	Checking of breaker closing and opening time	B	
	Measurement of contact resistance of the individual module and complete board	B	
	Testing of CTs, PTs etc.	B	
6.17.7	Protection scheme and logics		
	Testing of numerical relay to verify setting and logics. Testing of all other electromechanical/static relay as per installation. Checking of all lockout relays.	B	
	Verify all the inputs & outputs (binary/analogue) are healthy	B	
	Check all changeover logics (verify with time log in SOE)	B	
	Check the final setting of relays as per master record	B	
	Checking of reverse blocking scheme in HT switchgear	B	
Note	Check the breaker operation on test / local / remote Trip the breaker on protection Check the final setting of relays	B B B	
6.17.8	Breaker cubicle		
	i) Free movement of aux.	B	
	ii) Free shutter movement	B	
	iv) Continuity of heater circuit & IR (thermostat setting)	B	
	v) Secondary wiring & fuses (Check for sulfurization if any) Physical condition, cleanliness, lubrication	B	
	vi) Test all relays on the cubicle (attach relevant test records)	A	Random by FQA
	vii) All excess openings are blanked	B	
	viii) Panel pressurization in service (if applicable)	B	
	ix) Panel identification levels provided on both front & back	B	

6.18	E.S.P.	A-FQA, B-EMD & BMD	
6.18.1	Physical condition of ESP parts: internal cleanliness, inspection for damaged / erosion of emitting and collecting electrodes, cleaning of gas distribution screen	B	
6.18.2	Check for spacing of collection plates, collecting plate to emitting wire spacing / alignment, distance between bottom of screen and front cable	B	Random check by A
6.18.3	Visual inspection for checking the snapping / damages of electrode and this replacement	B	Random check by A
	i) Spark rate, leakage through manhole to be checked and corrected ii) ESP field healthiness, no. of rapping mechanism working etc needs to be checked iii) All the valves and fittings to be checked for body, bonnet, seat leakage, local/remote operation to be checked, cleanliness of gauge glass to be ensured	B B B	
6.18.4	Gas distribution tests check healthiness of GD screen, replace wherever necessary, if possible, carry on. Structural integrity and positioning of all the diverter plates on gas distribution screen may be ensured	A+B	
6.18.5	Review of rapping frequency, if required	B	

6.19.	BOTTOM ASH HOPPER	A-FQA B- AHP/BMD	
6.19.1	Check for condition of refractory in the trough and peep hole of the BA hopper. Replace the refractory with new one wherever refractory is damaged	B+A	Refractory replacing process and drying period to be checked by FQA
6.19.2	Check condition of flushing / quenching nozzles and replace wherever required	B	AHP
6.19.3	Check the condition of structural members of the hopper including plates and take necessary action as required	B+A	FQA+AHP
6.19.4	Check the condition of Clinker grinders & rollers, feed gates & associated equipment	B	AHP
6.19.5	Check the condition of Wire mesh inside & replace the damaged portion	B	BMD
6.20	DRY ASH SYSTEM / FLY ASH SYSTEM	A-FQA B-AHP	
6.20.1	Transport Air System		
	Inspection of Coupling bushes / alignment checking & safety / relief valve servicing. Cleaning of air filters / moisture traps	B	
6.20.2	Dust Conditioners		
	Clean spray nozzles, oil replacement, Check & replacement of chain/rubber seals, if required	B	
6.20.3	Rotary feeders		
	Cleaning of seal air pipes, gear box oil replacement, Realignment of feeder, if required	B	
6.20.4	Silo Fluidizing Blowers / Buffer Hopper Blowers		
	Cleaning of suction filters, tighten / replace V-belts, oil replacement, Servicing the safety valve	B	
6.20.5	Silo vent fans / Scavenger fans		
	Check the condition & tightness of belts / replace, if required realignment & suction damper servicing	B	
6.20.6	Wash water pumps / Dust condition water pumps / Drain pumps		
	Gland packing replacement, alignment & bearing checking	B	
6.20.7	Vent filters		
	Inspection/Cleaning/replacement of bag, filters, cleaning & replacement of aeration pads	B	
6.20.8	Buffer Hoppers		
	Inspection / Cleaning / replacement of bag filters	B	
6.20.9	Vacuum pumps		
	Check/inspect belt tightness & replace the belts, if required bearings, greasing & replacement of gland packing	B	
6.20.10	Wetting Unit		
	Vacuum pump suction pipe chokage clearing,	B	

	Wetting head & Cyclone air washer nozzle cleaning, check the condition of DRC lining impact plate & collector tank tail piece replacement, if required		
6.21	ASH SLURRY PUMP	A-FQA B-AHP	
6.21.1	Dismantle the pump for servicing, check clearances and change worn out parts	B	
6.21.2	NDT on pump shaft surface to check for surface defect and take necessary action as required	A+B	
6.21.3	Reassemble the pump check, clearances and correct as required	B	
6.21.4	Check the condition of gland packing replace the parts in case of damage and reinstate and check for leakage	B	
6.21.5	Check condition of fluid coupling, bushing and bolts. Replace the damaged parts and retighten the bolts to specified torque value	B	
6.21.6	i) Align the pump and couple with the drive / in case of V belt tightness to be checked, condition of belt and pulley to be checked and replace if required ii) Check for seal water pressure iii) Trial run of the pump for one hr. in recirculation where recirculation is available	B B	
6.21.7	Visual inspection of the full length of the ash slurry pipeline, check the fixing of pipe support clamps and tighten the loose one, replace the damaged clamps	B	
6.22	ASH SLURRY PIPELINE	A-FQA B-AHP	
6.22.1	Random thickness measurement of the pipe by Ultrasonic D. meter. In case of reduction of thickness beyond limit on one side, rotate all such pipes by 90 deg	A	
6.22.2	In case of reduction in thickness in the entire periphery of pipe beyond limit, all such pipes to be replaced and ends to be welded using proper procedure and qualified welder. To ensure the quality of the weld random radiography to be done and take n.a. in case of defect	B	
6.23	ASH WATER PUMP	A-FQA B-AHP	
6.23.1	Dismantle the pump for servicing, check clearances and change worn out parts	B	
6.23.2	Reassemble the pump check, clearances and correct as required	B	
6.23.3	Check the condition of gland packing replace the parts in case of damage and reinstate and check for leakage	B	
6.23.4	Check condition of bushing and bolts. Replace the damaged parts and retighten the bolts to specified torque value		
6.23.5	Trial run of the pump for one hr.	B	

6.24	CW PUMP AND SUMP	A-FQA B-MMOS	
6.24.1	CW PUMP i) Dismantle the pump for servicing ii) Clean and inspect various parts i.e. impeller, casing sleeves, guide pieces etc. for wear. Replace the damaged parts iii) Clean and inspect bearings, Polish / replace damaged bearings and sleeves iv) Check the condition of column pipe and weld in the pipe, DPT to be conducted on suspected area v) Assemble pumps, check and adjust the clearances on pump bearings, impeller / casing and guide pieces etc. vi) Check coupling condition, change the damaged parts, align with the drive and take trial run.	B B B A B B	
6.24.2	CW SUMP & RELATED EQPMT i) Remove lock gates, check the rubber seals, clean and repair, paint if required with proper quality paint and check the ii) Remove filters, clean and repair damaged wire mesh, paint with suitable paint as required. iii) CW sump and other pumps sump cleaning work (cleaning of sludge/foreign material etc.)	B B B	
6.25	SEA WATER PUMP & SUMP	A-FQA B-MMOS	
6.25.1	SEA WATER PUMP i) Dismantle the pump for servicing ii) Clean and inspect various parts i.e. impeller, casing sleeves, guide pieces etc. for wear. Replace the damaged parts iii) Clean and inspect bearings, Polish / replace damaged bearings and sleeves iv) Assemble pumps, check and adjust the clearances on pump bearings, impeller / casing and guide pieces etc. v) Check coupling condition, change the damaged parts, align with the drive and take trial run	B B B B B	
6.25.2	SEA WATER SUMP & RELATED EQPMT.		
	i) Travelling water screen-1&2: Cleaning / Checking for mesh damage & rectification, Nozzles proper positioning, Tightening of bolts etc.	B	
6.26	SWEET WATER PUMP & SUMP	A-FQA B-MMOS	

6.26.1	SWEET WATER PUMP i) Dismantle the pump for servicing ii) Clean and inspect various parts i.e. impeller, casing sleeves, wear rings etc. for wear, Replace the damaged parts iii) Clean and inspect bearings, Polish / replace damaged bearings and sleeves iv) Assemble pumps, check and adjust the clearances on pump bearings, impeller / casing and wear rings, etc. v) Check coupling condition, change the damaged parts, align with the drive and take	B B B B B	
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	trial run		
6.26.2	SWEET WATER SUMP & RELATED EQPMT i) Remove lock gates, check the rubber seals, clean and repair, paint if required with proper quality paint and check the paint thickness ii) Sweet water sump and other pumps sump cleaning, work (cleaning of sludge/foreign material etc.)	B B	
6.27	CONTROL AND INSTRUMENTATION	A-FQA B-C&I	
6.27.1	Transmitters i) Visual Inspection of root valves, isolation valves, manifolds, impulse lines, drain pots, for any leakages ii) Visual inspection of terminals of transmitter and local JB's, connecting wires, conduits, cable trays, transmitter covers. iii) Check for calibration iv) Check for zero in case of flow transmitters	B B B B	
6.27.2	Pressure and Flow Switches		
	i) Visual inspection of root valves, isolation valves, manifolds, impulse lines, drain pots, for any leakages	B	
	ii) Visual inspection of terminals of pressure and flow switches and local JB's, connecting wires, conduits, cable trays, pressure and flow switches cover	B	
	iii) Check for switch settings i.e. set & reset values	B	
6.27.3	Temperature Switches		
	i) Check for switch settings i.e set & reset values	B	
	ii) Visual inspection of terminals of temperature switch and local JB's, connecting wire, conduits, cable trays and temperature switch covers	B	
6.27.4	Pressure Gauges		
	i) Visual Inspection of root valves, isolation valves, manifolds, impulse lines, drain pots, for any leakages	B	
	ii) Visual inspection of fills in gauges for transparency, leakages	B	
	iii) Visual inspection of front glass covers for transparency breakages	B	
	iv) Check for calibration	B	
6.27.5	RTD's and Thermocouples		
	i) Visual inspection of thermo wells and temperature elements	B	
	ii) Visual inspection of heads of the temperature elements	B	
	iii) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable	B	

	trays		
	iv) Check for calibration as per schedule	B	
6.27.6	Level Switches		
	i) Visual Inspection of root valves, float body	B	
	ii) Visual Inspection of switch assemblies	B	
	iii) Visual Inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	iv) Check for level switch settings	B	
6.27.7	E/P Converters		
	i) Visual Inspection of air lines in and out of E/P Converter	B	
	ii) Visual Inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	iii) Visual Inspection of covers for dust ingress etc.	B	
	iv) Check for calibration	B	
6.27.8	Positioners		
	i) Visual Inspection of air lines	B	
	ii) Visual Inspection of covers for dust ingress etc.	B	
	iii) Check for calibration	B	
	iv) Check for the limit switch operation and carry out adjustments if necessary	B	
	v) Cleaning of flapper nozzle with petrol (proper care to be taken when using petrol)	B	
6.27.9	Pressure Regulators		
	i) Visual Inspection of air lines	B	
	ii) Visual Inspection of air filters and cleaning wherever applicable	B	
	iii) Check for settings	B	
	iv) Check for pressure gauge healthiness as well as its calibration	B	
6.27.10	Pneumatic Actuators		
	i) Visual Inspection of all pneumatic lines, joints, diaphragm etc. for leakages	B	
	ii) Checking of actuator manually for full stroke of valve operation and adjustments if any required	B	
	iii) Checking of actuator with pneumatic signals for full stroke of valve operation	B	
	iv) Checking of limit switches operation and adjustments, if any,	B	
6.27.11	Power Cylinders		
	i) Inspection of all pneumatic lines, joints, power cylinder etc. for leakages	B	
	ii) Checking of actuator with pneumatic signals for full stroke of valve operation	B	
	iii) Checking of limit switches operation and adjustments, if any.	B	
	iv) Checking for interlock s, if any.	B	
6.27.12	Motorized Valve Actuators		
	i) Visual Inspection of physical condition, cleanliness, grease/actuator gear oil condition	B	

	ii) Cleaning of limit switch assembly as needed	B	
	iii) Check for leakages	B	
	iv) Checking of limit / torque switch during operation adjust, if required	B	
	v) Checking of limit / torque feedbacks	B	
6.27.13	Position Transmitters		
	i) Visual Inspection of position transmitter and its mechanical links	B	
	ii) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	iii) Check for calibration	B	
6.27.14	Panel instruments including indicators / recorders		
	i) Check for physical condition	B	
	ii) Zero checks	B	
	iii) Check for calibration	B	
6.27.15	Servo Units		
	i) Check the condition of the oil, for moisture, dirt etc.	B	Chemistry-analysis of oil sample
	ii) Oil top-up / filling as per procedures	B	
	iii) Visual inspection of oil lines of servo unit	B	
	iv) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	v) Check for the pressure of oil & adjust if necessary	B	
	vi) Cleaning of associated oil filters	B	
	vii) Check for the full stroke operation of the valve	B	
	viii) Check for associated limit switch operation		
	ix) Check for calibration	B	
6.27.16	Speed Pickups of Feeders		
	i) Visual inspection of speed pickup and its mechanical links	B	
	ii) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	iii) Cleaning of pick up and check for dust ingress if any		
	iv) Check for outputs	B	
6.27.17	Load Cells of Feeders		
	i) Visual inspection of load cell and its mechanical links	B	
	ii) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable trays	B	
	iii) Check for healthiness of outputs	B	
	iv) Carry out calibration of feeder	B	
6.27.18	Cooling Fans & Filters		
	i) Check for noise and vibration	B	
	ii) Visual inspection of connectors, JB's, terminals, connecting wires, conduits, cable	B	

	trays		
	iii) Cleaning of fan blades	B	
	iv) Cleaning of filters	B	
	v) Check for lubrication if required	B	
6.27.19	Power Supplies		
	i) Check for cooling fan, any abnormal temperature, voltage level, load etc.	B	
	ii) Cleaning of all internals, terminals & cooling fans etc.	B	
	iii) Checking for any leaky electrolytic capacitors, battery back up etc. and replacements if required	B	
	iii) Checking for rated load for definite period and check for temperature etc.	B	
6.27.20	Vibration Pick-up		
	i) Check for healthiness of pickup as per O&M Manuals	B	
	ii) Calibrate vibration pickup as per manufacturer's recommendations	B	
6.27.21	Differential Expansion Pick-up		
	i) Check for healthiness of pick up as per O&M Manual	B	
	ii) Calibrate vibration pickup as per manufacturer's recommendations	A+B	
6.27.22	Shaft Vibration pick-ups		
	i) Check for healthiness of pick up as per O&M Manual	B	
	ii) Calibrate Eccentricity pickup as per manufacturer's recommendations	B	
6.27.23	Axial Shift Pick-ups		
	i) Check for healthiness of pick up as per O&M Manual	B	
	ii) Calibrate Eccentricity pickup as per manufacturer's recommendations	A+B	
6.27.24	Eccentricity Pick-ups		
	i) Check for healthiness of pick up as per O&M Manual	B	
	ii) Calibrate Eccentricity pickup as per manufacturer's recommendations	B	
6.27.25	Speed Pick-ups		
	i) Check for healthiness of pick up as per O&M Manual	B	
	ii) Calibrate Eccentricity pickup as per manufacturer's recommendations	B	
6.27.26	Electrohydraulic Governor		
	i) Calibrate EHC as per manufacturer's recommendations	A+B	
	ii) Calibration of all governing rack instruments	B	
	iii) Cleaning & servicing of EHG	B	
	iv) Checking resistance of trip solenoid & maintain record	B	
	v) Operation checking of Turbine Trip Solenoid	A+B	

	vi) Servicing & checking of limit switches	B	
	vii) Checking of ATT instruments as per OIN	B	
	viii) Operation checking of all control valves & calibration of feedback transmitters	B	
	ix) Cleaning all junction boxes related to governing system & terminal tightness checking	B	
6.27.27	Turbine Stress Evaluator		
	i) Check for healthiness of TSE thermocouples, CJC, RTD's etc.	B	
	ii) Check for the different curves & LVM's as per O&M manual	B	
	iii) Check for the voltage levels in individual power supplies and for any abnormal temperatures	B	

6.27.28	Turbine & Boiler Protections		
	i) Check for calibration of all field devices / actuating elements	A+B	Random by FQA
	ii) Check for each protection channel and ensure its healthiness	A+B	Random by FQA
	iii) Check for LSR of Governing system & load Transducer	B	Recommissioning group/OPN/EMD/C&I
6.27.29	Instrument Air Line		
	i) Visual inspection of leakage especially at joints, flanges etc.	B	
	ii) Check for any abnormal sound / noise etc. for any passing on line	B	
	iii) Carry out air drop tests for all instrument lines for detection of leakage	A+B	Random by FQA
6.27.30	Uninterruptible Power Supply		
	i) Check for any abnormal sound, voltage, temperature etc.	B	
	ii) Cleaning of all electronic cards, panels, thyristors, cooling fans, contactors, etc.	B	
	iii) Check for electrolyte level, Sp. Gravity etc. (if applicable)	B	
	iv) Check for battery conditions like voltage, capacity, connecting leads, cable joints, etc.	A+B	
	v) Charging of batteries	B	
	vi) Check for inverter, charger & static switch setting as per OEM recommendations	B	
	vii) Check for changeover, circulating current (in case of parallel inverters)	A+B	
6.27.31	24 VDC Power Supply		
	i) Check for any abnormal sound, voltage, temperature etc.	B	
	ii) Cleaning of all electronic cards, panels, thyristors, cooling fans, contactors, etc.	B	
	iii) Check for electrolyte level, Sp. Gravity etc. (if applicable)	B	
	iv) Check for battery conditions like voltage, capacity, connecting leads, cable joints, etc.	A+B	
	v) Check for settings as per OEM recommendations	B	
	vi) Charging of batteries	B	
	vii) Check for changeovers	A+B	

6.27.32	Grounding		
	i) Panel earthing, Instrument Earthing, Power supply earthing to be checked as per OGN COS-ISO-00-OGN/OPS/C&I/003 Rev No:02 with earth resistance clamp-on meter at panel end	B	
	ii) All DCS panels including Network panel iii) All Third-party panel or NON DCS panels like TSI, PDB, MTM fieldbus, etc.		
	iv) All Field installed panels or PDBs Charger, UPS and DCDB and ACDB		
	v) Check Earthing and Keep record of Earth grid or Earth PIT	B	
	vi) Identification of earth risers, High emphasis on VMS system risers	B	
	vii) Isolation to be ensured for electronic shield grounding and panel grounding (By measuring between panel earth and signal earth)	B	
	viii) Ensure grounding scheme of the C&I system is followed as per the approved drawing/design	B	
	ix) Electronic Earth pit to be measured in every Unit O/H preferably by 3-point (fall-of-potential) method	B	
	x) Proper tightness of terminations at earth pit/grid raiser ends and terminations covered with grease or bitumen	B	
6.27.33	Power Supply Monitoring		
	i) Check all DCS and network panel power supply redundancy from the source end and record all the alarms availability	B	
	ii) Check all the Third-party Panel mounted in CER or Field* for Power supply redundancy from source and record availability of alarms. Third party panels must also include the Hydraulic system and its OSU power supply. iii) *All Third-party and Field Mounted panel (LPBP, HPBP LPBP OSU panel, LLD panel, TSI panel, MTM panel, Generator temp measurement panel, APH Fire detection panel, start-up system panel, LMU, PW, CG, MOT/CFT Temperature ctrl valve, Flame scanner, Hydrastep, Local PDBs, Relay panel supply and other local panel)	B	
	iv) Check availability of all sources from common DCDB and its redundancy checking in Offsite areas	B	
	v) Thermography monitoring of UPS, charger including ACDB and DCDB, CER and field panels	B	
	vi) Monitoring of alarms in Static switches installed in PDB and its supply changeover. Keep record of any fluctuation in DPUs, Networking switch restart, bad of Vibration or scanner o/p or panel desired performance.	B	

6.27.34	DDCMIS		
	i) Cleaning/ replacement of panel filters & cooling fans	B	
	ii) Sealing of glands & other opening	B	
	iii) Tightness checking of terminal blocks	B	
	iv) Controller redundancy checking changeover checking: Documentation required	B	
	v) Network /BUS redundancy checking. Documentation required.	B	
	vi) Checking of all alarms/SOEs related to change over/failure with protocol	B	
	vii) Checking for any simulation/forcing in the controller (removal of forcing / record keeping should be done)	B	
	viii) Software backup of the latest logic before and end of overhaul	B	
	ix) Checking the status of CMOS/DPU battery (if provided) & replacement if necessary	B	
	x) Check Hardware backup draw out panel healthiness with operation	B	
	xi) Third party or DCS panels MINI UPS changeover and battery healthiness, note down the backup time. xii) Replace battery if required.	B	
6.27.35	UPS		
	i) Cleaning of panel internals & filters	B	
	ii) Sealing of glands & other opening	B	
	iii) Tightness checking of terminal blocks & all connections	B	
	iv) Check all the capacitors, diodes, and thyristors for their healthiness (if available) Replace the deteriorated components	B	
	v) Checking and servicing all cooling fans (replace if required)	B	
	vi) Check and ensure that all fuses are as per design vii) rating and specification (follow replacement schedule)	B	
	viii) Check Battery healthiness by allowing load test on battery	B	
	ix) Check and record all settings (DIP switch, if any)	B	
	x) Checking of synch between both inverters & also changeover from main to alternate along with operation: Documentation required	B	

	xi) Checking of all related alarms at local & also at control room with operation: Documentation required	B	
	xii) Identification of source (if not done earlier) for main& alternate	B	
	xiii) Labelling of UPS & source feeder on UPS panel (if not done earlier)	B	
6.27.36	Charger		
	i) Cleaning of panel internals & filters	B	
	ii) Sealing of glands & other opening	B	
	iii) Tightness checking of terminal blocks & all connections	B	
	iv) Check all the capacitors, diodes and thyristors for v) their healthiness (if available). Replace the deteriorated components.	B	
	vi) Checking and servicing all cooling fans (replace if required)	B	
	vii) Check and ensure that all fuses are as per designrating and specification (follow replacement viii) schedule)	B	
	ix) Check Battery healthiness by allowing load test on battery	B	
	x) Check and record all settings (DIP switch, if any)	B	
	xi) Both charger on battery, note down the survival time till low voltage along with operation	B	
	xii) Checking of all related alarms at local & also at control room along with operation. Documentation xiii) required.	B	
	xiv) Identification of source (if not done earlier) for main& alternate	B	
	xv) Labelling of chargers & source feeder on charger control panel (if not done earlier)	B	
	xvi) Ensure proper grounding of panels & negative (if provided)	B	
6.27.37	HPBP /LPBP		
	i) Check NAS and TAN value before start of servicing and keep record Cleaning of tank by cambric cloth or markin cloth after emptying the entire system oil (replace oil as recommended by OEM)	B	
	ii) Check oil pump interlocks	A+B	

	iii) Check for any oil leakages	B	
	iv) Replace filters	B	
	v) Cleaning & servicing of servo valves and cylinders hydraulic actuators	B	
	vi) Calibrate pressure transmitters/gauges	B	
	vii) Checking of accumulator N2. pressure and top-up if required Keep record of date of filling	B	
	viii) Check of relief valve	B	
	ix) Checking functionality of fast opening device	B	
	x) Keep record of Pump ON-OFF timing if applicable or xi) Accumulator charging Valve On-OFF timing and keep record Check for any deterioration	B	
	xii) Calibration of actuators & feedback transmitters and stroke checking and its operation with operation executives. Check Full closure of valves with xiii) operation and TMD. Check Valve Open Close timing.	A+B	
	xiv) Interlock & protections checking with Operation and EEMG	B	
6.27.38	TSI System / Aux Drives		
	i) Note down the TSI readings while turbine is on barring gear & standstill condition after unit shutdown	B	
	ii) Isolate the power supply to TSI panels	B	
	iii) Proper labelling & storage of sensors associated clamps & fittings to be ensured during removal & restoration. Note down the individual gap voltage of sensors before removing of any sensors	B B	
	iv) Calibration protocols to be maintained with linearity check	B	
	v) Ensure proper signal shielding for all sensors and avoid any ground loop	B	
	vi) Checking of cable route (should not laid with HT cable nearby)	B	
	vii) While fixing sensors proper tightness must be ensured	B	
	viii) Ensuring "zero" reading with equipment not in service	B	
6.27.39	LLD		
	i) Cleaning, servicing of LLD, Replace the gasket / O-rings	B	
	ii) Calibration of LLD and joint protocol signed with operation	B	

6.27.40	Servers / EWS / OWS / LVS		
	i) Before shutdown of the machine keep latest backup / image files	B	
	ii) Shutdown the SERVER/EXS/OWS one by one	B	
	iii) Clean the system with blower/ vacuum cleaner	B	
	iv) Powering on of the system	B	
	v) Ensure healthiness of portable UPS (if provide with the system)	B	
	vi) Note down the battery backup time of portable UPS (Battery may be replaced if required)	B	
	vii) Ensure latest antivirus running on the system	B	
	viii) Asset list updating and verification as per OGN OGN-OPS-C&I-004 Security Policies for Plant Control Systems	B	
	ix) Disabling of USB ports in servers	B	
	x) Check changeover of Server and historian: monitor any data loss and HMI loss	B	
6.27.41	Scanner		
	i) Isolation of scanner panel after unit shutdown	B	
	ii) Removal of all scanners from field, before boilerwater wash	B	
	iii) Cleaning and checking of scanner guide pipe assembly (replace if required)	B	
	iv) Checking of alignment of guide pipe and mounting of scanner as per drawing inside the guide pipe	B	
	v) Checking of scanner cooling air hose (must be fixed properly to scanner guide pipe)	B	
	vi) Calibration of scanner as per OEM O&M manual: keep the record	B	
	vii) Checking of scanner cables, connectors & cable route (Not touching hot surfaces) Checking of Shield connections as per drawing	B	
	viii) After running of scanner air fan air flow to individual scanner must be ensured	B	
	ix) Cleaning of scanner control panel Ensuring availability of redundant power supply & proper earthing	B	
	x) Cleaning, servicing of all scanners and its calibration as per O&M manual	B	
	xi) Functionality checking of scanner with torch light after panel charging	B	
	xii) Functionality checking of 3/4. voting card	B	
6.27.42	Igniters		
	i) Isolation of power supply to HEA igniters & transformers	B	
	ii) Removal of igniters	B	

	iii) Cleaning & Servicing of igniters	B	
	iv) Servicing of advance/retract power cylinders	B	
	v) Servicing of solenoids, limit switches	B	
	vi) Servicing of air filter regulators	B	
	vii) Operation checking of power cylinders & limit switches	B	
	viii) Cleaning all junction boxes related to igniters	B	
	ix) Spark checking in lab with igniter test kit: kept therecord	B	
	x) While mounting the igniters in field length adjustment should be done by physically inspecting inside the boiler. Keep the record.	B	
6.27.43	Pneumatic Actuators		
	i) Removal of Air filter regulator, air lock relay, positioner, EP converter & PFT (if available) from the actuator	B	
	ii) Cleaning & servicing of above instruments at Lab	B	
	iii) Servicing of limit switches	B	
	iv) Mounting of above instruments to the actuator	B	
	v) Check for any air leakage, jamming of valve etc.	B	
	vi) Check functionality of manual hand wheel & neutral position	B	
	vii) Adjust the limit switches	B	
	viii) Calibration of pneumatic actuator (note down the opening & closing time)	B	
	ix) Check functionality of air lock by reducing the air pressure	B	
	x) If necessary, painting of actuators may be done	B	
	xi) Remote checking of actuator	B	
6.27.44	Field instruments		
	i) 100% calibration of all P&I and CLCS instruments, other instruments and maintain record as per IMS standard	B	
	ii) Cleaning of junction boxes, terminal tightness at JB& instruments	B	
	iii) Checking for healthiness of temperature duplex elements and check TT is configured for dual changeover element	B	
	iv) CJC healthiness checking and attend defect	B	
	v) Dynavane filter controller checking	B	
	vi) Check cable routing (not touching hot zones or surfaces)	B	
	vii) Checking, inspection and purging of Annubar	B	

	viii) MDBFP and Dosing Scoop calibration	B	
	ix) Rotate the positions by 180 deg for elements mounted in flue gas path to minimise erosion	B	
6.27.45	Analysers (SWAS, CEMS, O2, etc.)		
	i) Cleaning of sensors, filters and drain lines	B	
	ii) Preservation of sensors in case of SWAS	B	
	iii) Checking and attending of Sample line for corrosion, chocking and leakages Check Heat tracing cable healthiness and replace if required	B	
	iv) Mounting of probe, probe position may be rotated to minimize erosion due to flue gas	B	
	v) Checking power supply & earthing of all instruments	B	
	vi) Sealing of flange peripheral to avoid air ingress	B	
	vii) Checking sample root valves & may be replaced by Mech maint if necessary	B	
	viii) Calibration of analyser and maintain record SWAS record to be maintained with chemistry	B	
	ix) Note down the cell constant & slope	B	
6.27.46	Acromat / MTM / Water Wall temperatures		
	i) Prepare the list of faulty and doubtful Acromat /MTM	B	
	ii) Check for routing, proper insertion in Pads and any damage of Acromat / MTM and replace if required check the doubtful Acromat / MTM with operation	B	
	iii) Healthiness and correspondence checking with Heat source (Eg: hot air gun) if not done during Commissioning, Protocol to be signed with operation*	B	
	iv) Clean check cable routing for Water wall temperatures measurement Check healthiness and availability/Healthiness of Duplex element till JB	B	
6.27.47	Blade Pitch actuator		
	i) Servicing of blade pitch actuators	B	
	ii) Sealing and canopy preparation of actuators	B	
	iii) Calibration and operation checking of actuator	B	
6.27.48	APH		
	i) Healthiness checking of thermocouple and thermowell 180 deg rotation of thermowell in flue gas path	B	
	ii) Rotor stoppage limit switch checking to be done	B	
	iii) Air motor solenoid healthiness checking/servicing to be done and maintain record with operation	B	
6.27.49	ASLD		

	i) ASLD servicing and calibration	B	
	ii) Checking of ASLD with Operation	B	
6.27.50	LVS	B	
	i) LVS projector cleaning	B	
	ii) Attending defects if any	B	
6.27.51	AHP		
	i) ALI calibration and checking, maintain record	B	
	ii) MHV cylinder Maintenance (Checking Passing & Repairing faulty cylinder)	B	
	iii) BH Pulse Jet Servicing	B	
	iv) Bottom ash local panel, instruments and drives checking, instruments and drives checking	B	
	v) UPS & charger checking as per main plant charger and UPS procedure: maintain record	B	
	vi) Mini UPS battery backup testing to be done. Maintain record.	B	
	vii) JBs Cleaning and attending instrument air leakages	B	
6.27.52	Junction box / LIE / LIR	B	
	i) Cleaning of junction boxes with air blower /service air	B	
	ii) Terminal tightness checking	B	
	iii) Sealing of gland holes (if left blank)	B	
	iv) Dressing of spare cables	B	
	v) Checking for proper closing /sealing of JB doors	B	
	vi) Replace liners/seal of door if required	B	
	vii) Ensure proper labelling of junction box/LIE/LIR	B	
	viii) Ensure red colour marking if related to protection	B	
	ix) If required painting of junction box/LIE/LIR may be done	B	
6.27.53	Aerofoil / Impulse piping for Air & Flue Gas measurements		
	i) Isolation /removal of instruments from mounting	B	
	ii) Physical inspection of aerofoil elements	B	
	iii) Cleaning of aerofoil holes	B	
	iv) Draining of accumulated dust from drain points of aerofoil	B	
	v) Cleaning of impulse lines with service air	B	
	vi) For flue gas related impulse lines pocking & cleaning of tapping points must be done	B	
	vii) Ensuring availability of continuous purge lines & setting purge air flow through rota meters	B	
6.27.54	Compressor		
	i) P&I checking		Opn/C&I/MM/EMD
	ii) Servicing of drain solenoids and dryer solenoids	B	

	iii) Healthiness checking of all instrument's calibration of Pressure/DP switches if applicable	B	
	iv) Coil healthiness checking of all solenoids	B	
	v) Availability of UPS supply to PLC	B	
	vi) Receiver tank drain solenoids servicing and checking	B	
6.28	CHEMICAL PLANT	A-FQA B-CHEM MM-OS	
6.28.1	WATER TREATMENT PLANT		
	Clarifier		
	i) (Actuators, partial plumes, mixing tanks) Thorough cleaning, dislodging algae removal, checking of rotary equipment for corrosion, painting and other maintenance of bridge Frequency: once in one year preferably post monsoon	B	
	ii) Alum lime and coagulant aid tanks, pumps and agitators tanks bucket and lining to be checked for any damage. Pumps suction strainers and recirculation lines should be cleaned and leakages to be checked. Check smooth running of stirrer and gear box overhaul. Checking of bearings. Painting of steel structures. Frequency: Cleaning once in six months and painting once in a year	B	
	iii) Gravity filter / ACF Checking of filter media height and top up in vessel. Mud balls and lumps of filter media, channeling of filter bed should be removed and screened and graded. Painting / lining of vessel should be checked for erosion / corrosion of internals and vessel. Checking of distributor, strainers or perforated pipes and header leakages. Doing of open / extended backwash etc. Rota meter cleaning Frequency: Once in two years or earlier if required	B	
	iv) Degasser Towers and Degasser tanks Cleaning of degasser storage tanks, checking for corrosion and inner lining of tanks. Checking proper functioning of float and indicating equipment. Frequency: Once in an year during overhaul Checking of inlet and outlet air pressure. Blower filter cleaning/replacement. Airblower efficiency and degasser efficiency. Degasser tower rascheling rings top up and arrangement, checking of water distributor header and spray arrangement, corrosion of lining of tower, condition of gas outlet hood. Frequency: Lining checking once in a year other once in two months	For tank lining A&B and for rest B	
	v) Cation / Anion / M.B. Vessels Checking of resin height and top up in vessel, rubber lining of vessel, condition of lateral headers and strainers. Tightness of header and laterals and analyzer probe condition and arrangement. Open backwash removal of dirt accumulation. Resin treatment if required with 10% NaCl and 2% NaOH. D.P. against resin trap to be checked. Frequency: Once in a year	B	Chemistry

	vi) Regenerant tanks, agitators, pumps All pumps and ejectors should be checked for proper dosing and flow, corrosion / pitting and gland condition. Any leakage must be checked. Tanks agitator and lining should be checked for corrosion and damage. Painting etc. draining arrangement of gland leakages and coupling guards Frequency: Once in an year	A+B	
	vii) Alkali / Acid / DM storage tanks One by one tanks should be checked for proper condition of rubber lining and corrosion, gauge glass or float condition and cleaning. Working of fumes absorbers. Vessel and lines painting, joint bolt corrosion and replacement. Frequency: Once in an year. Valves diaphragms, CO2 absorbers and absorbent replacement and level checking of absorbent Frequency: Once in two months	A+B	
	viii) CPU Regeneration Acid / Alkali / Dosing Tanks: One by one tanks should be checked for proper condition of rubber lining and corrosion, gauge glass or float condition and cleaning. Vessel and lines painting, joint bolt corrosion and replacement All pumps should be checked for proper dosing and flow, corrosion / pitting and gland condition. Any leakage must be checked. Draining arrangement of gland leakages and coupling guards. Frequency: Once in an year	B	
	ix) Chlorination plants: Overhauling and replacement of corroded parts and lines. Proper painting and cleaning of system valves diaphragms and control equipment checking. Any leakage should be attended immediately. Frequency: As and when required. Overhaul once in a year.	B	
	x) Chlorine cylinders Check for proper functioning of valves, corrosion of cylinder body and painting with standard colour code. Pressure test certificate must be renewed regularly Frequency: As and when needed CLAS(Chlorine leak absorption system): Checking of air headers, checking of caustic strength and level in the tank, Proper	B	Chemistry
	functioning of Pumps, blowers and Chlorine leak detectors Frequency: Min once in a month, joint checking to be done for routine checks of CLAS	B	Chem and C&I
	xi) ETP/Neutralization Pit/Coal Slurry Settling Plant a) Cleaning of Tube settling plant, Flash mixer Tanks, Alum & Lime dosing tanks & checking for corrosion & any damage to coating & painting, if required b) Cleaning of Neutralization pit & checking for any damage to coating & painting, if required. c) CSSP pit cleaning & checking for any damage to coating & painting, if required.	B	
6.29	SEA WATER ELECTRO CHLORINATION PLANT	A-FQA B-MM-OS, EMD, C&I	

6.29.1	Overhauling and replacement of corroded parts and lines. Proper painting and cleaning of system valves & diaphragms. Any leakage should be attended immediately. All pumps, blowers should be checked for proper dosing and flow, corrosion / pitting and gland condition. Any leakage must be checked. Draining arrangement of gland leakages and coupling guards. Vessel and lines painting, joint bolt corrosion and replacement Hypo Chlorines Storage Tank: Cleaning & Checking for any damaged coating Frequency: As and when required. Overhaul once in a year	B	
6.29.2	Electrolyser	B	
	Cleaning of salt deposits on electronic cards & inside the panel. Tightness checking of transformer, choke & other components, check for any corrosion & painting, if required. Ensuring input/output voltage & current		
6.30	MAIN PLANT	A-FQA B-MM-TG+MM-PP+OPN+CHEM	
		Frequency: With overhaul	
6.30.1	Turbine blades for deposition: Collection and analysis of Turbine depositsamples	B	
6.30.2	Condenser:		
	i) Steam side Condenser tubes : - Excessive wear at baffles. Impingement from turbine. Oil and / or oxide on surfaces and corrosion in specific areas of Condenser shell. Tube erosion from DM make up	B	
	ii) Water side Algae, fungi, scale, tube end corrosion, water box corrosion, tube corrosion / impingement, and deposition	B	
6.30.3	Heat exchanger Debris / mud scale		
	i) Steam side : Oil, oxide deposits, exfoliation	B	
	ii) Water side Oil, oxides, pitting	B	
6.30.4	De-aerator		
	Oxide loose debris, inspection for pitting colour of steam and waterside in storage tanks	B	
6.30.5	Economizer Oxide debris	B	
6.30.6	i) Boiler drum Loose oxides, tightness of cyclones and their position, damage to cyclone separator, surface of drum and colour of steam and water side of drums	B	
	ii) Headers Pitting, other deposits	B	
	iii) Furnace tube Oxide thickness corrosion sites pitting	B	
	iv) Superheater tube Deposits	B	
	v) Reheater tubes Deposits	B	
6.30.7	Evaporators water side Scale build up corrosion	B	

6.30.8	Lubricating oil pipe work Main oil tank oil Centrifuge deposits, filter / strainerexamination	B	
6.31	ADDITIONAL CHECKS IN TG AREA	A-FQAB-TMD	
6.31.1	Other important checks in TG i) Seal bore dimensions check ii) LPT compensator assembly--- iii) Expansion bellows DPT iv) LP rotor blades locking plate -DPT v) Root weld joints of Guide blades—Visual and DPT vi) Servo motor assembly-Cone Disc, Springs etc –DPT vii) HP-IP, IP-LP and LP-GEN couplingbolts –Visual and DPT viii) HP inner, IP inner and outer StudsDPT &UT ix) Stop and Control valves –ConeSpindle etc --DPT x) MS &HRH strainers –Visual and DPT xi) U-seal rings—Visual and DPT xii) JOP line –Visual and DPT xiii) LP exhaust spray line—Visual andDPT xiv) Feed storage tank –Visual and DPT xv) HP heater—Visual and DPT	A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B A+B	
6.31.2	GENERATOR	A-FQA B-EMD	
	Other Important Checks i) Exciter fan hub keyway—Visual& DPT ii) Rotor Compressor blades-Stationaryand moving- Fluorescent DPT iii) Retaining rings –UT and DPT iv) Seal oil and Primary water pipelineweld joints---Visual &DPT v) H2 Gas system weld joints—Visualand DPT	A+B A+B A+B A+B A+B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.1	Conveyer belt replacement 1. Check for weak belt joint. 2. Check for belt condition (any cut, edge damage, air pockets etc.). 3. Check for joint alignment. 4. Check for scrapper adjustment. 5. Check for proper belt tension. 6. Check for coupling tightness. 7. Check for proper skirt rubber positions. 8. Check for all idlers are in place. 9. Check for deck plate & seal plate in position. 10. Check for foreign material & removal of the same, especially return side of belt and tail pulley area. 11. Check for belt sway (during trial run) and rubbing of edges and adjustment / training. 12. Check all coupling and tail pulley guard are in place. 13 Check for proper take-up pulley / trolley and associated rope pulley/ sheave/diversion movement (during trial run).	B B A B B B B B B B B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.2	Replacement of Pulley/Gearbox/Hydraulic couplings 1. Check for proper bolt tightness 2. Check for proper oil/lubricant filling up to recommended level 3. Check for proper oil / lubricant grade 4. Check for pulley lagging condition 5. Check for proper coupling bolt tightness and lubrication (in case of gear coupling) 6. Check for Alignment between motor-fluid coupling-gear box-pulley 7. Check for any abnormal sound (during trial run) 8. Check for excessive heat generation from bearing housing (during trial run) 9. Check for vibration within specified limit (during trial run) 10. Check for proper oil pressure (in case of AOP provided in Fluid coupling) 11. Check for proper oil cooling (in case of scoop type of Fluid coupling) 12. Check for any oil / cooling water leakages	B B B B B A B B B B B	
6.32.3	Drive alignment (HT) 1. Check for alignment reading (within specified limit) 2. Check for coupling bolt tightness 3. Check for soft foot 4. Check for any abnormal sound (during trial run) 5. Check for excessive heat generation from bearing housing (during trial run) Check for vibration within specified limit (during trial run)	A B B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.4	Wagon Tippler Replacement of gearbox/fluid coupling/Hydraulic pump/Hydraulic motor/hydraulic cylinders/ clamp assembly/rack & pinion/drive shafts/drive shaft bearing/hyd system oil flushing/Side beam/Structures/Counterweight/drive support Pedestal foundation works:		
	1. Check for the proper drive alignment. 2. Check for coupling bolt tightness 3. Check for proper alignment of matching parts. 4. Check for any abnormal sound (during trial run). 5. Check for excessive heat generation from bearing housing (during trial run). 6. Check for vibration within specified limit (during trial run). 7. Check for proper picking of line / flushing of line / tanks before hydraulic oil / lubricating oil filling (in case of complete replacement of lines). 8. Check for proper oil/lubricant filling up to recommended level. 9. Check for proper oil / lubricant grade. 10. Check for quality of oil/lubricant. 11. Check for any oil leakages from gearbox/fluid coupling/hydraulic cylinder/hydraulic pump/hydraulic motor. 12. Check for oil leakages form hose / tubes. 13. Check for the integrity of jointed/ coupled part / structure. 14. Check for the strength of structure. 15. Check for total counterweight. 16. Check for proper / smooth turn table movement (during trial run). 17. Check for proper clamping of wagon. 18. Check for lubrication of pedestal / Plummer block. 19. Check for proper parking of turn table. 20. Check for proper functioning of various limit switches. 21. Check whether all civil foundations are intact.	A B B B B B B B B B B B B A A B B B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.5	Wagon tippler: Replacement of oil cooler assembly/ oil purifier assembly/ oil tank cleaning/ Hopper lining/gunniting works/ Rail alignment 1. Check for the proper drive alignment. 2. Check for coupling bolt tightness. 3. Check for proper alignment of matching parts. 4. Check for any abnormal sound (during trial run). 5. Check for excessive heat generation from bearing housing (during trial run). 6. Check for vibration within specified limit (during trial run). 7. Check for proper picking of line / flushing of line / tanks before hydraulic oil / lubricating oil filling 8. Check for proper oil/lubricant filling up to recommended level. 9. Check for proper oil / lubricant grade. 10. Check for quality of oil/lubricant. 11. Check for any oil leakages from gearbox/fluid coupling/hydraulic cylinder/hydraulic pump/hydraulic motor. 12. Check for oil leakages form hose / tubes. 13. Check for the integrity of jointed/ coupled part / structure. 14. Check for proper function of various limit switches 15. Check whether all civil foundations are intact	A B B B B B B B B B B B B B A B A	
6.32.6	Stacker Reclaimer: Replacement of slew bearing/bucket wheel and slew drive gear box/hydraulic cylinder (Boom luffing)/rack & pinion/Bucket wheel assembly/Bucket wheel bearings/shrink discs/Slew bearing greasing system line flushing/structures/counterweight		

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.6.1	1. Check for the proper drive alignment 2. Check for Check for coupling bolt tightness 3. Check for proper alignment of matching parts 4. Check for water level of slew bearing (during Repl) 5. Check for any abnormal sound (during trial run) 6. Check for excessive heat generation from bearing housing (during trial run) 7. Check for proper torque tightness of shrink disc 8. Check for proper grease supply in all the lubrication points of slew bearing / plummer blocks / travel wheels 9. Check for proper counter weight balancing and hydraulic cylinder pressure setting (with the help of experts) 10. Check for integrity of structure 11. Check for the proper / smooth functioning of boom luffing 12. Check for proper picking of line / flushing of line / tanks before hydraulic oil / lubricating oil filling 13. Check for proper oil/lubricant filling up to recommended level 14. Check for proper oil / lubricant grade 15. Check for quality of oil/lubricant 16. Check for any oil leakages from gearbox/fluid coupling/hydraulic cylinder/hydraulic pump/hydraulic motor 17. Check for oil leakages form hose / tubes SR:Replacement of gearbox/fluid coupling/Hydraulic pump/Hydraulic motor/hydraulic cylinders /hyd rail clamp/drive shafts/buckets/clutch assembly/ thruster brake assembly/travel wheels/rail alignment/replacement/hyd system line flushing/cleaning/ replacement of oil cooler/oil purifier/CRD assembly 1. Check for the proper drive alignment 2. Check for Check for coupling bolt tightness	A B B B B B B B B A B B B B B B B B B B A B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
	3. Check for proper alignment of matching parts 4. Check for proper functioning of slew drive clutch 5. Check for any abnormal sound (during trial run) 6. Check for excessive heat generation from bearing housing (during trial run) 7. Check for proper torque tightness of shrink disc 8. Check for proper grease supply in all the lubrication points of slew bearing / plummer blocks / travel wheels 9. Check for proper counter weight balancing and hydraulic cylinder pressure setting (with the help of experts) 10. Check for integrity of structure 11. Check for the proper functioning of thruster brakes / rail clamp 12. Check for proper picking of line / flushing of line / tanks before hydraulic oil / lubricating oil filling 13. Check for proper oil/lubricant filling up to recommended level 14. Check for proper oil / lubricant grade 15. Check for quality of oil/lubricant 16. Check for any oil leakages from gearbox/fluid coupling/hydraulic cylinder/hydraulic pump/hydraulic motor 17. Check for oil leakages form hose / tubes 18. Check for proper function of CRD	B B B B B A B B B B B B B	
6.32.7	CRUSHER Drive alignment/Rotor assembly balancing 1. Check for the proper drive alignment 2. Check for Check for coupling bolt tightness 3. Check for proper alignment of matching parts 4. Check for Rotor balancing 5. Check for vibration within limit	A B B B B	
6.32.7.1	CRUSHER: Replacement of gearbox/fluid coupling/rotor assembly/rotor shaft/ full set hammer /rotor bearings/ Cage frame assembly/spring deck works		

[illegible]

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.9	Paddle Feeder Replacement of rotor assembly/ rotor blade full set/gear box/fluid coupling/hydraulic pump/hydraulic motor/oil cooler/oil purifier/CRD assembly/hyd system flushing/cleaning/ rail replacement/alignment/drive alignment 1. Check for the proper drive alignment 2. Check for Check for coupling bolt tightness 3. Check for proper alignment of matching parts 4. Check for any abnormal sound (during trial run) 5. Check for excessive heat generation from bearing housing (during trial run) 6. Check for proper function of CRD 7. Check for the proper functioning of thruster brakes 8. Check for proper picking of line / flushing of line / tanks before hydraulic oil / lubricating oil filling 9. Check for proper oil/lubricant filling up to recommended level 10. Check for proper oil / lubricant grade 11. Check for any oil leakages from gearbox/fluid coupling/hydraulic cylinder/hydraulic pump/hydraulic motor 12. Check for oil leakages form hose / tubes 13. Check for proper function of oil cooler / oil filter 14. Check for rail alignment	A B B B B B B B B B B B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.10	Mobile Trippers Replacement of gearbox/rail clamp assembly/thruster brake assembly/drive shaft/ travel wheels/CRD assembly/ drive alignment/ rail replacement/alignment 1. Check for the proper drive alignment 2. Check for Check for coupling bolt tightness 3. Check for proper alignment of matching parts 4. Check for any abnormal sound (during trial run) 5. Check for excessive heat generation from bearing housing (during trial run) 6. Check for proper rail alignment 7. Check for proper CRD/ Festooning functioning 8. Check for proper flap gate functioning	A B B B B B B B	
6.32.11	Belt Feeders Replacement of gearbox/fluid coupling/rack & pinion/Belt/Pulley 1. Check for proper bolt tightness 2. Check for proper oil/lubricant filling up to recommended level 3. Check for proper oil / lubricant grade 4. Check for quality of oil/lubricant 5. Check for pulley lagging condition 6. Check for proper coupling bolt tightness and lubrication (in case of gear coupling) 8. Check for Alignment between motor-fluid coupling-gear box-pulley 9. Check for any abnormal sound (during trial run) 10. Check for excessive heat generation from bearing housing (during trial run) 11. Check for vibration within specified limit (during trial run)	B B B B B B A B B B	

S.NO.	DESCRIPTION OF ITEM AND TYPE OF CHECK	CLASS OF CHECK	REMARKS
6.32.	COAL HANDLING PLANT	A-FQA B-CHP	
6.32.12	Conveyor drive/Equipment overhauling Final assembly 1. Check for proper bolt tightness 2. Check for proper oil/lubricant filling up to recommended level 3. Check for proper oil / lubricant grade 4. Check for quality of oil/lubricant 5. Check for pulley lagging condition 6. Check for proper coupling bolt tightness and lubrication (in case of gear coupling) 7. Check for all idlers in place 8. Check for Alignment between motor-fluid coupling-gear box-pulley 9. Check for any abnormal sound (during trial run) 10. Check for excessive heat generation from bearing housing (during trial run) 11.Check for vibration within specified limit (duringtrial run)	B B B B B B B A B B B	
6.33.	OFFSITES		
6.33.1	Cooling Towers Drift eliminators Nozzle Cleaning Replacement of Damaged nozzles Distribution Pipes & End Pipes Cold Basin Steel & concrete structure Fan Deck Tower Casing Fibre/Plastic Structure Railings, rung ladder, stairs & walkways Hot water duct for Cross flow tower & its thickness measurement Hoist & Lifting arrangement CT Outlet Bucket Strainer Peripheral drains Fan Gear Boxes Drive Couplings Fan Stack Stack door Lightning Arresters Vibro Switches Oil Temp Guages Stop log Gates Vegetation & Debris around CT area Aviation Lighting	B B B B B A A A A A A A A A A A B B B B B B B B B B A B B B B B B	

	Splash Fills & Film fills Algae deposition on CT internals Fan Blade tip clearance		
6.33.2	CW Pump Overhaul Bluematch with Sole plate for VT Pumps Alignment of Pump & Motor Clearance b/n Sleeve & Bush Lift setting Inspection of Pump REJ & change it if required Inspection of Filters, cleaning & replacement of filter elements in lube water system Inspection of Vortices breaker: Floor Cone, Splitter Plates, Wall fillets etc. if available Level Measurement by Straight edge Shaft Run out Shaft UT Shaft NDT:DPT & MPT of used ones Ratchet Assembly if applicable Thrust Bearing Inspection Ensure Proper locking of Impeller Wear ring clearances Shaft Keyways & Muff 2/2 rubber bearing inspection incase of CVP Impeller Inspection for Wear & Pitting Stop Log gates Lube Water Filters Lube Water System Valves Lube water Tank & Protection REJ	A B B B B B B B B A A B B B B B B B B B B B	
6.33.3	CW Discharge Butterfly Valve Valve Seal Inspection Hydraulic Pump Cut-in & cut-off Valve Bearings Valve seat clamping ring Valve Opening Time Valve Closing Time: Normal & Emergency Disc Locking Arrangement in Open & Close position Hard Barrication near Valve Counter Weight area Gear Box inspection for Motorized Valve Valve Freeness Valve Positioner:Open/Close Marking Valve Operation check by smart actuator	B B B B B B B B B B B	
6.33.4	CW Duct Inspection Complete dewatering of Supply & Return ducts 100% complete inspection of ducts	B	

	Repair coating/painting for CW ducts internal surface	B	
	Thickness measurement of above ground ducts	B	
	Servicing of Air Release Valves and its isolating valves	B	
	Ensuring Confined space permit, work at height permit inside ducts	B	
	Painting of above ground duct portion	B	
	Removal of debris from ducts	B	
	Fitness of Divers to be ensured	B	
	All essential PTW's to be ensured	B	
	Check the healthiness of CW interconnection valve and replace the seal if required	B	
	Inspection of Pitot tube pits and its associated system	B	
	Checks for coating, painting & thickness measurement	A	
	Joint protocol for final inspection of ducts before charging by Offsites & Operation	B	

6.34 : UNIT OVERHAULING NORMS FOR NTPC THERMAL UNITS

Norms as prescribed through various IOMs & O&M manuals have been adopted as required and mentioned below.

Generator inspection:

The recommended norm to perform initial inspection of generator as per O&M manual is 10,000 Hrs < T (equivalent operating hours) < 20,000 Hrs.

For Subsequent interval between two major inspections the Equivalent Operating Hours norm is 40,000 Hrs < T equiv. < 60,000 Hrs.

Generator Inspection/Overhauling may be carried out every 4 years.

Boiler overhauling:

The present practice to carry out boiler overhauling is once in two years.

Criteria for OH – Boiler Running hours >14000hrs or 3 years from last OH whichever is earlier.

Turbine overhauling:

LPT insitu PAUT & MPI – Every 2 years

LP turbine overhaul to carry out in every 4 years.

HP/IP to carry out in every 6 years.

OH types:

Type E: Boiler & Aux, Turbine Bearings. Insp. (if reqd.) for every 2 years.

Type F: Boiler & Aux, LP Turbine Cylinder O/H for every 2 years.

Type G: Boiler & Aux, HP/IP/LP Turbine Cylinder O/H for every 6 years.

Ref. 5 – IOM – REF.NO.:OS/ST/OH/ Dt.27.02.2018

O/H Rolling plan need to followed as per revised norms as below

SL NO	TURBINE TYPE	FROM MARCH 2016 ONWARDS	FROM 2018 ONWARDS
1	500 MW –KWU HP+IP+LP (New Fleet-Simhadri Stg-1 & Stg-2)	LP- Every 4 years HP / IP- Every 8 Years IOM: OS/ST/ OH/ 9526, Dt 16.03.2016	LP- Every 2 years HP / IP- Every 8 Years Ref IOM: OS/ST/ , Dt 27.02.2018

Ref. 4 – IOM – REF.NO.:OS/ST/OH/19526 Dt.16.03.2016

O/H Rolling plan need to followed as per revised norms as below

SL NO	TURBINE TYPE	FROM 2012 ONWARDS	FROM MARCH 2016 ONWARDS
1	500 MW –KWU HP+IP+LP (New Fleet-Simhadri Stg-1 & Stg-2)	HP /IP /LP – Every 6 years IOM: OS/ST/MT/OH/1369,Dt 25.10.2012	LP- Every 4 years HP / IP- Every 8 Years IOM: OS/ST/ OH/ 9526, Dt 16.03.2016

Ref. 3 – IOM – REF.NO.:OS/ST/MT/OH/1369 Dt.25.10.2012

O/H Rolling plan need to followed as per revised norms as below

SL NO	TURBINE TYPE	FROM 2005-06	FROM 2012 ONWARDS
1	500 MW –KWU HP+IP+LP (New Fleet-Simhadri Stg-1 & Stg-2	LP- Every 4 years HP / IP- Every 8 years	HP /IP /LP – Every 6 years IOM:OS/ST/MT/OH/1369,Dt 25.10.2012

Ref. 2 – IOM – REF.NO.:OS/ST/MT/Catenary/Rev01/3189 Dt.10.01.2014

Main Turbine Catenary Checking & Correction during First Inspection

1. Catenary correction along with bearing inspection, & decoupling of HP-IP and IP-LP etc should not be done during first inspection of Generator in case machine is running with normal turbovisory parameters.
2. However, if there is a problem of high vibration or barring gear stalling, the work scope to rectify the problems of these machines shall be finalized with CC-OS-ST group separately.

Ref. 1- D(O)'s review on Overhauling practices – 5/12/2000

S.NO.	UNIT CAPACITY	MAIN AREAS IDENTIFIED FOR O/H/INSP	TYPE OF O/H	PERIOD OF O / H
1.	500 MW (KWU/GEC)	- Boiler & Aux. - Turbine Brg. Insp. (if reqd.)	E*	30 days
2.	500 MW (KWU/GEC)	- Boiler & Aux. - LP Turbine Cylinder O/H - Generator Inspection / repair - Generator Transformer	F	35 days
3.	500 MW (KWU/GEC)	- Boiler & Aux. - IP/HP Turbine Cylinder O/H - Generator Inspection / repair - LP Turbine Inspection	G	45 days

Note : 1. * On 5/12/2000 in “D(O)’s review of overhauling practices” it was agreed to carryout Boiler overhaul for 25 days in case of 500 MW units.

2. To control the O&M cost following are the guidelines :
 - a) Sequence of O/H for the station should be so adjusted that not more than two major O/H are taken up in a year.
 - b) For ensuring proper quality of work during O/H, O/H of two units at a time to be avoided
3. If overhauling of any unit at a station is altered from its planned schedule then prior information regarding the same should be sent to D(O) and OS.

7.0 **IMPLEMENTATION & RESPONSIBILITY CENTRES**

Responsibility for the class of checks in Maintenance Quality Plan is as follows :

<u>Class of Check</u>	<u>Responsibility</u>
A	To be witnessed by FQA
B	To be witnessed by O&M
A+B or B+A	To be jointly witnessed by O&M and FQA

Quality checks protocols are to be prepared by contractor / O&M section concerned and jointly to be signed by contractor, O&M executive and FQA executive as applicable.

The following procedure is to be followed for preparation of quality plan and protocol for overhauling/maintenance and release of SES as per requirement.

ACTIVITY	RESPONSIBILITY
Preparation of quality plan by maintenance section concerned.	Maintenance Section concerned
Verifying and approval of the quality plan.	FQA
Creating PR in SAP with material group “service-mechanical-FQA/” or electrical or C&I as the case may be, and attaching the Quality Plan with PR . (This will enable FQA to release SES through SAP)	Maintenance section concerned.
Attaching Quality Plan to the PO.	FQA
Generation of overhauling / maintenance order through SAP.	Maintenance section concerned & MTP
Execution of work through maintenance order, preparation and signing of protocol as required by quality plan (applicable class of checking).	Maintenance section concerned & FQA.
Attachment of the protocol to the maintenance order.	Maintenance sections concerned.
Generation of SES through the maintenance order and release by Engineer. Submitting SES for release by FQA.	Maintenance section concerned
Release of SES by FQA and issuance of quality certificate. (Protocol attached to the maintenance order can be viewed from the maintenance order- ML81N >SES)	FQA
Release of SES by EIC and submitting for payment.	Maintenance section concerned.

8.0 EXCEPTION REPORTING

- Exceptions if any shall be recorded & maintained by FQA as per CQA approved procedure vide QS-01-FQA-P-12.
- All O&M responsibility centers shall record the exceptions & maintain the record at their end. One copy of the report may be sent to MTP.

NTPC		EXCEPTION REPORT		SIMHADRI	
TITLE: QUALITY MANAGEMENT IN MAINTENANCE / OVERHAULING					
LMI/OGN/OPS/SYST/018			REV: Number, Month ‘Year		
Unit No.	Description of exception	Location of exception		Remarks	

9.0 RECORD KEEPING

The records to be maintained under this LMI include

- Protocols of all checks, exception reports & compliance reports at respective O&M sections concerned. Copy of the same may be maintained at a centralized location at MTP.
- Exception reports put up to HOP to be maintained at FQA Dept.

10.0 REVIEW

This document shall be reviewed by HOP- Simhadri on two yearly basis.

ANNEXURE

NTPC	COMPLIANCE RECORD	SIMHADRI
TITLE : QUALITY MANAGEMENT IN MAINTENANCE / OVERHAULING		
REF NO: LMI/OGN/OPS/SYST/018		

SR NO	DESCRIPTION OF THE ACTIVITY AND TYPE OF CHECK	COMPLIANCE

Annexure-A: UNIT OVERHAULING NORMS FOR NTPC THERMAL UNITS

Ref. 1. D(O)'s review on Overhauling practices – 5/12/2000
 2. OS/O/H/ 25/03/97.

SL. NO.	UNIT CAPACITY	MAIN AREAS IDENTIFIED FOR O/H / INSP.	PERIOD OF OVERHAULING
1.	200 MW / 210 MW (KWU & LMW)	- Boiler & Aux. - Turb. Brg. Insp. (if required)	25 days
2.	200 MW / 210 MW (KWU/GE)	- Boiler & Aux. - LP Turbine Cylinder O/H - Generator Inspection / Repair - Generator Transformer	35 days
3.	200 MW / 210 MW (KWU/GE)	- Boiler & Aux. - IP / HP Turbine Cylinders O/H - Generator Inspection / Repair - LP Turbine Inspection	40 days
4.	200 MW /210 MW (LMW)	- Boiler & Aux. - All the Turbine Cylinders O/H - Generator Transformer	45 days
5.	250/500 MW (KWU /GEC)	- Boiler & Aux. - Turbine Brg. Insp. (if required)	30 days
6.	250/500 MW (KWU/GEC)	- Boiler & Aux. - LP Turbine Cylinder O/H - Generator Inspection / repair - Generator Transformer	35 days
7	250/500 MW (KWU/GEC)	- Boiler & Aux. - IP/LP Turbine Cylinder O/H - Generator Inspection / repair - Generator Transformer	40 days
8	250/500 MW (KWU /GEC)	- Boiler & Aux. - IP / HP Turbine Cylinders O/H - Generator Inspection / Repair - LP Turbine Inspection	45 days
9	660 MW/800 MW	- Boiler & Aux. - Turb. Brg. Insp. (if required)	35/40 days

10	660 MW/800 MW	- Boiler & Aux. - Generator Inspection / Repair - LP Turbine Inspection	45 days
11	660 MW/800 MW	- Boiler & Aux. - HP/IP/LP Turbine Cylinders O/H - Generator Inspection / Repair	60 days

**Annexure-B: Revised Rolling plan for capital overhauling of Steam Turbine
200/500 MW units (As per IOM dated 09/11/20)**

S.N	Turbine type	Description
1	210 MW-KWU, HP+IP+LP (Old fleet)	All bearings Inspection, MOP inspection & General OH works & all turbine valves – Every 2 years HP/IP – Every 6 years or 50, 000 Running Hrs LP- 4 yrs or 30,000 Running Hrs RLA – Every 6 yrs or 50,000 Running Hrs
2	LMZ 200/210MW, HP+IP+LP (Old fleet)	All Bearings Inspection, MOP inspection & General OH works & all Turbine valves – Every 2 years HP/IP – Every 6 years or 50, 000 Running Hrs LP- 4 yrs or 30,000 Running Hrs RLA – Every 6 yrs or 50, 000 Running Hrs
3	250/195 MW KWU (New fleet- Bongaigaon, BRBCL, Kanti, Barauni, Bhilai)	1 st Complete overhaul of Bearings, MOP inspection & General OH works, LP, Turbine valves, barring gear hydro motor, Governing system- 1 Year (along with 1 st Generator Inspection) Next all bearings Inspection, MOP inspection & General OH works & all Turbine valves, barring gear hydromotor – Every 2 years Next LP- Every 4 years or 30,000 Running hours 1 st HP, IP – 6 Years or 50,000 Running hours Next HP, IP – Every 8 years or 64,000 Running Hrs RLA – 1 st : 1 lakhs running hrs & subsequent Every 8yrs
4	500MW – KWU, HP+IP+LP Old fleet (Ver 01 & 02)	All Bearings Inspection, MOP inspection & General OH works, all Turbine valves – Every 2 years LPT insitu PAUT & MPI – Every 2 years
5	500MW – KWU, HP+IP+LP New fleet (Ver 03 & above)	LP – Every 4 years or 30,000 running Hrs. HP/IP – Every 6 years or 50,000 Running hrs RLA – 1 st : 1 lakhs running Hrs & subsequent Every 6yrs or 50,000 Running Hrs
6	200 MW GE/ 500MW GEC HP+IP+LP	All Bearings Inspection, General OH works, all Turbine valves – Every 4 years HP/IP/both LP – Every 6 years or 50,000 running Hrs RLA – Every 6yrs or 50,000 running Hrs
7	500MW, ABB HP+IP+L P	All Bearings Inspection, General OH works , all Turbine valves – Every 4 years LPT- Every 4 Yrs or 30,000 Running Hrs HP/IP – Every 8 years or 64000 running Hrs RLA – Every 8 Hrs or 64,000 Running Hrs
8	200MW, Ansaldo HP+IP+LP	Every 6 years
9	110MW, Skoda HP+IP+LP	All Bearings Inspection, General OH works , all Turbine valves – Every 2 years HP/IP/LP – Every 6 years or 50,000 running Hrs RLA – Every 6 yrs
10	62.5MW and below	Every 5 years
11	Drive Turbine of TDBFPs	Bearing inspection & train alignment, MCV/ACV Servomotor: every 2yrs All make & design every 6 years along with complete governing system

Note:

- Running Hours means equivalent running hours considering cold/hot/warm start-ups.
- The above rolling plan is to be adhered, however the same will be reviewed again with station during OH scope of work finalization with COS-ST.
- Generator rotor & exciter CC-bolt gas tightness checking and seal ring inspection shall be done during intervening LP turbine overhaul of 500 MW KWU Turbine if Generator inspection is not envisaged in Scope of Work.
- Boiler – Boiler overhauling is carried out with a time gap of 2 years. Criteria for OH - Running hours
>14000hrs or 3 years from last OH whichever is earlier.

Annexure-C: Rolling plan for OH/servicing of TG & Aux. equipment's during OH.

Sl No.	Equipment	Frequency	Work to be taken up during Overhaul	Compliance Report
1.	Replacement of compensator	>1.5 lacs of running hours	To be done during OH.	
2.	Replacement of disc springs for Stop & control v/v servomotors	>1 lac of running hours	To be done during OH.	
3.	Replacement of turbine casing parting plane fasteners.	>1 lac of running hours	To be done during OH.	
4.	HP turbine Inlet & casing U-seal ring replacement	>1 lac of running hours	To be done during OH.	
5.	MS & HRH strainers replacement	>1 lac of running hours	To be done during OH.	
6.	OIL tanks	Every OH : Two Year	Main Oil tank, TDBFP Oil tank cleaning should be done	
7.	CW butterfly valve	Every OH : Two Year	CW inlet & Outlet valves, ACW/ECW pumps & PHE B/F V/V servicing/door seal replacement to be done to facilitate condenser pass cleaning during unit running condition. door seal replacement.	
8.	All critical drain, vent, warm up lines: Material Mix-up checking, Thickness survey, DPT of weld joints.	Every OH : Two Year	Material Mix-up & Thickness survey of all critical drain lines like MS, HRH, HPBP & CRH drain lines, Feed water vent/drain lines	
9.	Vacuum pump servicing	Every OH : Two Year	Vacuum pumps servicing to be done and heat exchanger to be cleaned.	
10.	Vacuum coolers cleaning	Every OH : Two Year		
11.	Vacuum Pump suction line NRV installation/Inspection	Every OH : Two Year	NRV need to be installed in suction line as per COS circular & Inspection of NRV's to be done during Unit OH.	

12.	Condenser steam chest inspection	Every OH : Two Year	1. Condenser steam chest inspection to be carried out for ensure healthiness of nozzles provided in DM make up & CEP recirculation lines. 2. Inspection of tie rod/supporting structure/pipes/joints NDT inspection and bellow protection pipes inspection	
13.	Piping Hangers inspection	Every OH : Two Year	All critical Piping (MS, HRH, CRH & FW) hangers needs to be inspected and adjusted	
14.	Deaerator Weld Joint NDT inspection	Every OH : Two Year	Deaerator weld joints NDT, Nozzles inspection, deaerator vessel thickness survey and continuous vent orifice inspection etc to be done as per LMI.	
15.	Deaerator nozzles inspection, Deaerator vessel thickness survey for erosion	Every OH : Two Year		
16.	Deaerator vent orifice size inspection	Every OH : Two Year		
17.	Seal steam header before & after drain thoroughness & Bypass orifice	Every OH : Two Year	Seal steam header before & after drain thoroughness & Bypass orifice inspection to stop moist steam.	
18.	Accumulators pressure checking & CF system healthiness checking	Every OH : Two Year	All accumulators & CF system healthiness checking with continuous vent / Orifice cleaning as per COS circular. Accumulators to be checked in all available opportunity also.	
19.	Control Fluid Pumps change over checking & Holding time measurement	Every OH : Two Year	Checking as per COS guidelines	
20.	Erection of vent line discharge of CF Pumps		Installation of vent line of individual control fluid pump	

21.	Control fluid tank cleaning	Every Four Year	CF tank cleaning in every four years	
22.	Checking of Main trip valve settings	Every OH : Two Year	Trip setting of Main Trip valves to be checked	
23.	Calibration of Low Vacuum trip device settings	Every OH : Two Year	Calibration of Low Vacuum trip device to be checked	
24.	LPT diaphragm inspection	Every OH : Two Year	Both LPT diaphragm healthiness inspection	
25.	COLTCS	Every OH : Two Year	Servicing of COLTCS as per OGN	
26.	Critical piping impulse line DPT	Every OH : Two Year	All critical drain lines & impulse line joints DPT upto first isolating valve to be done.	
27.	CPU vessel servicing, resin trap Inspection and inlet,outlet & bypass valve servicing	Every OH : Two Year	Inspection of inlet, outlet and bypass valve servicing and resin traps and rubber lining hardness measurement	
28.	CRH NRV Position feedback	Every OH : Two Year	Proper mounting of position feed back of CRH NRV	
29.	CRH NRV Overhauling	Every OH : Two Year	Overhauling of CRH NRV as per COS Guidelines	
30.	TDBFP ACV charge with CRH	Every OH : Two Year	Servicing of ACV & MCV servomotors	
31.	hydrogen & primary water Control valve servicing	Every OH : Two Year	Servicing of hydrogen and primary water temp control valve and inspection of working nut	
32.	Condenser flood test	Every OH : Two Year	As per guidelines of COS circular	
33.	MOT coolers cleaning	If Lub oil temp is greater than 48 deg at any time of Year	MOT tube bundle cleaning (Oil side) after removal of cooler from position	
34.	Inspection of Rubber expansion joints of CW inlet and outlet line	Every OH : Two Year	Inspection of healthiness of RE Bellow	

35.	Inspection of Flash tanks from inside	Every OH : Two Year	Inspection of Flash tanks from inside to ensure healthiness of nozzles, spray nozzles. Thickness survey of flash tanks and connected manifold connection to be done	
36.	Inspection of Compensators of Flash tanks	Every OH : Two Year	NDT inspection of flash tank bellows to be done.	
37.	MDBFP hydraulic coupling Overhauling	3 months	Oil properties checking, filter vent checking, greasing of scoop tube bearing.	
38.		Every OH: Two year	Checking of coupling, actuator, fusible plus, scoop tube assy., magnetic center, coupling gap etc.	
39.		1st: After 10 yrs. 2nd: 6 yrs after 1st OH 3rd: Every 4 yrs thereafter	Overhauling of MDBFP hydraulic coupling as per OEM	
40.	Inspection of butterfly valves of MOT vapor exhauster system and seal steam system	Every OH : Two Year	Inspection of butterfly valve to be done	
41.	Replacement of BFP Cartridges	As per EEMG recommendations		
42.	Overhauling of Drive Turbines	Every 06 years	Overhauling of Drive turbines to be done as per COS Guidelines	
43.	Overhauling of TG Valves including extraction NRV, HPBP & LPBP valves	Every OH : Two Year	Overhauling of TG Valves to be done as per COS Guidelines	
44.	Overhauling of Governing system	Every Capital OH	Overhauling of Governing system	
45.	MOT Mulsifier checking	Every OH : Two Year	Sealing of MOT after completion of MOT Mulsifier Checking	
46.	Servicing of AOP's & JOP's and ensure healthiness of JOP system PRV & Safety Valve	Every OH : Two Year	Servicing of AOP and jacking oil system. Ensure proper functioning of PRV and safety valves	
47.	Cleaning of all PHE's	Every OH : Two Year	Cleaning of all PHE	
48.	Replacement of TDBFP Exhaust valve door seal	Every OH : Two Year	Replacement of Door seal of TDBFP Exhaust valves	

49.	Replacement of TG valves servomotors disc springs	After 100000 running Hrs	Replacement of disc springs as per COS Guidelines	
50.	Hydro test of Lub Oil and jacking Oil system as per OD	After 150000 running Hrs	Hydro test of Lub Oil and jacking Oil system as per OD	
51.	Mapping of valves for following valves for passing 1. High energy drain valves 2. BFP RC valves 3. BFP discharge valve 4. BFP discharge NRV 5. HP heaters Isolation 6. FRS valve	Every OH : Two Year	Identify valves for passing before shutdown of machines. Rectify the defects.	
52.	Servicing of extraction NRV's	Every OH: Two year	Servicing of Extraction NRV to be carried out to ensure healthiness	
53.	Condenser tube eddy current testing	SS Tubes: After 10 years, Every COH/ recommendation of earlier test (whichever is earlier) Cu-Ni: Every COH	Measurement of thickness of Condenser tubes to ensure healthiness	
54.	Seal steam header drain before manual valve open to atmosphere		Manual valve open to atmosphere before seal steam station needs to be provided, if not provided.	
55.	GSC Drain Modification		GSC drain line size increase with modified strainer and trap	
56.	HPBP anti rotation device installation		Installation of anti-rotation device as per COS circular	
57.	PHE bypass line		ECW outlet valve bypass line of one inch line to be provided for facilitate slow charging of PHE	
58.	Lub Oil filter modification		Installation of view glass and vent line modification as per RCA committee recommendations	
59.	Provision of alarm on first stage pressure >2 ksc with turbine in trip condition		Provision of alarm on first stage pressure >2 ksc with turbine in trip condition to be incorporated in system	

60.	Extraction compensators (GE make)	Opportunity S/D / RSD	Inspection & providing additional supports.	
61.	Rotation of Lub oil & control oil filter	Once in a week & during opportunity.	Rotation of filters & cleaning of element.	
62.	Condenser	During opportunity	Water box & tube cleaning during opportunity. Readiness of bullet/jet cleaning manpower & contract.	
63.	Trial run of AC/DC JOP	24 hours after unit outage/opportunity.	Trial run of AC/DC JOP as per circular.	
64.	FRF stock	Continuous	Emergency stock to be maintained as per identified stations.	
65.	Condenser vacuum system	Continuous	Periodic Helium leak test to address air ingress/ DO issues.	
66.	RLA of turbine components	During every COH after 20 yrs of operation/ as per last RLA recommendations.	To be done during OH.	

Annexure-D Overhauling Rolling plan for Generator for Coal Stations and Hydro (Koldam)

Generator OH Schedule				Interval in Year			
Model	Pop	Station	OEM name	1st	2nd	3rd	4th
800-Hitachi	2	Lara	Hitachi	1	4	4	4
800-Siemens	2	Gadarwara	Siemens/BHEL	1	6	6	6
	3	Patratu					
800-GE	2	Telangana	GE	2	4	4	4
800-Toshiba	3	Kudgi	Toshiba	2	4	4	4
	2	Darlipali					
660-Alstom	2	Solapur	Alstom/GE	2	4	4	4
660-GE	3	Nabinagar	Alstom/GE	2	4	4	4
660-GE	2	Tanda-II	GE	2	4	4	4
660-KWU BHEL	2	Barh-II	Siemens/BHEL	2	6	6	6
	3	North Karanpura					
	2	Mouda-II					
660-L&T MH	2	Khargone	L&T MH	1	4	4	4
660-Power Machines	3	Sipat-I	Power Machines	1	4	4	4
	3	Barh-I					
660-Toshiba	2	Meja	Toshiba	2	4	4	4
500-ABB	2	Talcher-K-I	ABB/GE	2	4	4	4
500-Alstom	2	Rihand-I	Alstom	2	4	4	4
500-KWU BHEL	48	Singrauli-II, Rihand-II&III, Unchahar-IV, Korba-II&III, Vindhyachal-II,III,IV&V, Sipat-II, Farakka-II&III, Kahalgaon-II, Talcher-K,-II Ramagundam-II,III Simhadri-I,II, Mouda-I, Jhajjar, Vallur, Dadri-II	Siemens/BHEL	2	4	4	4
250-BHEL	7	BRBCL/Bongaigaon	BHEL	2	4	4	4
210-Ansaldo	3	Ramagundam-I	Ansaldo	2	4	4	4
210-BHEL-THW	2	Unchahar-I	BHEL	2	4	4	4
210-BHEL-THW	4	Dadri-Coal-I	BHEL	2	4	4	4
210-BHEL-THRI	3	Unchahar-II & III	BHEL	2	4	4	4
210-Elec-A	6	Vindhyachal-I	Electosila	2	4	4	4
210-Elec-R	4	Kahalgaon-I	Electosila	2	4	4	4
200-BHEL-A-THW	5	Singrauli-I	BHEL	2	4	4	4
200-BHEL-HY	4	Koldam	BHEL	2	4	4	4
200-BHEL-THW	6	Farakka-I, Korba-I	BHEL	2	4	4	4
195-BHEL	2	Kanti-II	BHEL	2	4	4	4
110-BHEL	4	Tanda-I	BHEL	2	4	4	4

Annexure-E Record sheet for trial run of motor

Time	Starting current (Amp)	Speed (RPM) (Amp)	No load current	Bearing DE	Temp NDE	Bearing vib.		Engr. in charge	Remarks
						DE H V A	NDE H V A		